

## Review article

# Vibration-based condition monitoring and fault diagnosis of rotating machinery: Fundamentals, signal processing techniques, standards, and emerging trends

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## ABSTRACT

Vibration-based condition monitoring and fault diagnosis are crucial for the reliable and safe functioning of rotating machinery. In recent years, there has been growing interest in integrating advanced data-driven and intelligent techniques for fault diagnosis, resulting in a substantial and rapidly expanding body of research in this area. However, most existing review studies are limited in scope, focusing on specific fault types, methods, and application domains. In addition, these studies lack a comprehensive synthesis that bridges classical vibration analysis and modern intelligent diagnostic approaches, as well as Industry 4.0-enabled technologies. This review presents a state-of-the-art synthesis of vibration-based condition monitoring and fault diagnosis in rotating machinery. The review synthesizes 238 peer-reviewed publications, international standards, and authoritative technical resources. The review begins with the fundamental principles of vibration-based condition monitoring and analyzes common fault mechanisms and their corresponding characteristic vibration signatures. Vibration signal processing techniques and intelligent diagnosis methods for rotating machinery are then reviewed. From this detailed synthesis, practical implementation challenges and research challenges are identified. The review identifies emerging trends and future research directions, including physics-informed intelligent diagnosis, hybrid AI frameworks, and digital twin-enabled predictive maintenance. These insights provide valuable guidance for researchers and practitioners seeking to advance vibration-based fault diagnosis for rotating machinery.

## 1. Introduction

Rotating machines, such as electric motors, pumps, compressors, turbines, fans, and gearboxes, are widely used mechanical systems in today's industrial applications [1,2]. The operational stability of these machines is crucial for maintaining productivity, product quality, energy efficiency, operational safety, and overall economic performance. Failures can cause unplanned shutdowns, production losses, damage to secondary equipment, environmental incidents, and, in some cases, safety hazards [3]. Therefore, effective monitoring and maintenance strategies are essential to ensure reliable operation, minimize downtime, and enhance asset performance throughout the machinery lifecycle.

Nowadays, maintenance practices have shifted from reactive

methods to preventive, condition-based, and predictive strategies to improve equipment availability while reducing maintenance costs and operational disruptions [4]. In this context, condition monitoring has become a key aspect of modern maintenance systems for continuously assessing machine health and detecting faults early, before catastrophic failure occurs [5,6]. Among various techniques, vibration-based condition monitoring (VBCM) has become one of the most popular and effective methods for detecting and diagnosing faults in rotating machinery [7,8]. The widespread use of VBCM is due to its high sensitivity to mechanical and electromechanical faults, its non-intrusive nature, cost-effectiveness, and suitability for both continuous online monitoring and periodic inspections. Since vibration responses directly indicate machine dynamic behavior, vibration analysis provides valuable

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diagnostic information for early fault detection, fault classification, and predictive maintenance decision-making [9].

Vibration is a natural response of mechanical systems caused by forces from rotating parts. Common sources of vibration are mass unbalance, shaft misalignment, mechanical looseness, bearing defects, gear damage, frictional interactions, fluid-induced excitations, electromagnetic forces, and structural resonances [10,11]. These excitation factors act individually or simultaneously, resulting in complex vibration responses that require advanced signal processing and diagnostic techniques for accurate interpretation. Under normal operating conditions, each machine exhibits a stable vibration signature determined by its design, operating speed, load conditions, structural dynamics, and environmental influences. When faults occur, this signature is altered, leading to measurable changes in vibration amplitude, frequency content, phase characteristics, or waveform patterns [12].

Prior to this work, several review papers on VBCM and the fault diagnosis of rotating machinery have been published [7,9,13–17]. These studies have contributed significantly to specific areas, such as sensor technologies [14], bearing fault diagnosis [17], signal processing methods [9], machine learning algorithms [15], and particular industrial applications [16]. Some studies [18,19] provide reviews of vibration analysis, whereas others review fault diagnosis [3,20]. However, most existing reviews are limited to specific subdomains and do not provide a complete view of the entire vibration-based diagnostic workflow. Moreover, recent advances in Artificial Intelligence (AI), the Industrial Internet of Things (IIoT), digital twins, edge computing, and wireless sensing have transformed vibration monitoring from conventional fault detection to intelligent predictive maintenance systems. Despite this progress, to the author's knowledge, there is no dedicated review that connects classical vibration analysis methods with modern Industry 4.0 technologies within a unified analytical framework. Consequently, the relationships among vibration fundamentals, fault mechanisms, diagnostic techniques, industrial implementation, and emerging intelligent technologies remain insufficiently synthesized within a single comprehensive review.

Therefore, this review presents a comprehensive and critical synthesis of vibration-based condition monitoring and fault diagnosis methods for rotating machinery. The review adopts a unified analytical framework that systematically integrates vibration fundamentals, measurement systems, fault mechanisms, signal-processing techniques, intelligent diagnostic methods, international standards, practical implementation challenges, and emerging Industry 4.0 technologies. In addition, the review critically discusses practical implementation issues, current research challenges, emerging research trends, and future research directions to provide a comprehensive perspective on the evolution of modern vibration-based condition monitoring. To define the scope of the review and guide its synthesis, the following Research Questions (RQ) are formulated:

**RQ1:** What are the fundamental principles governing vibration generation, propagation, and measurement in rotating machinery?

**RQ2:** What characteristic vibration signatures are linked to common mechanical, fluid-induced, and electrical faults?

**RQ3:** How do vibration signal characteristics and diagnostic methods relate to fault mechanisms in rotating machinery?

**RQ4:** How do international standards and practical implementation practices affect the reliability of vibration-based condition monitoring systems?

**RQ5:** What emerging technologies and research directions are shaping the future of vibration-based condition monitoring in the context of Industry 4.0?

The rest of this review is organized as follows. Section 2 presents the review methodology and analytical framework adopted to guide the organization and synthesis of the literature. Section 3 details the fundamentals of vibration-based condition monitoring and measurement systems (RQ1). Section 4 examines fault mechanisms and their characteristic vibration signatures (RQ2). Section 5 reviews signal processing

techniques and intelligent diagnostic frameworks for rotating machinery (RQ3). Section 6 discusses international standards, industrial implementation practices, and emerging trends in vibration-based condition monitoring (RQ4 and RQ5). Finally, Section 7 concludes the review.

## 2. Review methodology

This section describes the review methodology and presents the analytical framework that guides the organization, analysis, and synthesis of the literature throughout the subsequent sections.

### 2.1. Data selection process

A comprehensive literature search was conducted using three major scientific databases: *Scopus*, *Web of Science*, and *IEEE Xplore*. These databases were selected because of their broad coverage of mechanical engineering, vibration analysis, signal processing, artificial intelligence, and condition monitoring research. In addition to peer-reviewed publications, relevant international standards and authoritative technical resources were consulted to support discussions related to vibration measurement, machine evaluation, and industrial implementation.

The literature search strategy was developed by combining major concepts related to vibration-based condition monitoring using the Boolean operators AND and OR, as presented in Table 1.

Peer-reviewed journal articles, review articles, conference papers, relevant book chapters, international standards, and authoritative technical publications published in English were considered. The literature selection followed the PRISMA guidelines [21]. Records identified from the selected databases were merged, screened according to pre-defined inclusion and exclusion criteria, and assessed through full-text review, while duplicate records, non-English publications, and studies outside the scope of vibration-based condition monitoring and rotating machinery were excluded. Backward snowballing of the reference lists was subsequently performed to identify additional relevant studies [22]. Following this systematic selection process, 238 documents were retained for qualitative synthesis.

### 2.2. Bibliographic and thematic analysis

A bibliographic analysis was performed to characterize the selected literature and provide a quantitative overview of research developments in vibration-based condition monitoring and fault diagnosis. The analysis examined publication trends over time, document types, and the thematic distribution of the selected studies.

**Table 1**  
Search terms used for literature identification.

| Keyword category      | Search terms                                                                                                                      |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Vibration monitoring  | ((“vibration-based condition monitoring” OR “vibration monitoring” OR “vibration analysis” OR “vibration signal analysis”)<br>AND |
| Rotating machinery    | (“rotating machinery” OR “rotating machine” OR “motor” OR “pump” OR “compressor” OR “turbine” OR “gearbox”)<br>AND                |
| Fault diagnosis       | (“fault diagnosis” OR “fault detection” OR “fault classification” OR “machine health assessment”)<br>AND                          |
| Signal processing     | (“FFT” OR “wavelet” OR “time-frequency analysis” OR “empirical mode decomposition” OR “envelope analysis”)<br>AND                 |
| Intelligent diagnosis | (“machine learning” OR “deep learning” OR “CNN” OR “transfer learning” OR “domain adaptation”)<br>AND                             |
| Industry 4.0          | (“IIoT” OR “digital twin” OR “edge computing” OR “predictive maintenance”))                                                       |

Figure 1 presents the annual publication trend of the selected literature from 2010 to 2026. The number of publications has increased steadily over the study period, reflecting the growing importance of vibration-based condition monitoring and fault diagnosis of rotating machinery and the sustained interest of both the research community and industry. It is important to note that a small number of earlier foundational references and undated international standards, technical resources, and website materials included in the review were excluded from the figure.

Figure 2 classifies the 238 selected documents by publication type into journal articles (including review articles), conference papers, book chapters, international standards, and technical reports/web resources. Journal articles constituted the largest proportion of the selected literature, accounting for 212 of the 238 publications. This indicates the substantial volume of experimental, methodological, and applied research in vibration-based condition monitoring and fault diagnosis.

To provide a quantitative overview of the review coverage, the selected studies were categorized according to their principal research themes, as presented in Table 2. Since some studies contribute to more than one research theme, the total number of thematic classifications exceeds the 238 documents included in the qualitative synthesis.

To provide a consistent analytical structure for the review, Figure 3 presents the analytical framework adopted in this study. The framework shows the logical progression from vibration fundamentals and measurement systems to fault mechanisms, signal processing techniques, intelligent diagnostic methods, international standards, practical implementation, and emerging Industry 4.0 technologies. It aligns the review structure with RQ1 to RQ5, thereby defining the analytical scope of each section and guiding the synthesis presented in the subsequent sections.

### 3. Fundamentals of vibration-based condition monitoring and measurement systems

This section presents the fundamental principles of vibration-based condition monitoring and measurement systems, addressing RQ1.

#### 3.1. Fundamentals of mechanical vibration

Vibration is the oscillatory motion of a body about its equilibrium or reference position. In mechanical systems, vibration occurs when components are subjected to time-varying forces that create periodic or nonperiodic motion around their steady-state positions [23]. From a dynamic perspective, vibration is the response of a mechanical system to internal or external excitatory forces. This response is governed by the system's mass, stiffness, and damping characteristics, as shown in Figure 4. Mass determines the resistance to acceleration, stiffness governs the resistance to deformation, and damping dissipates vibrational energy.

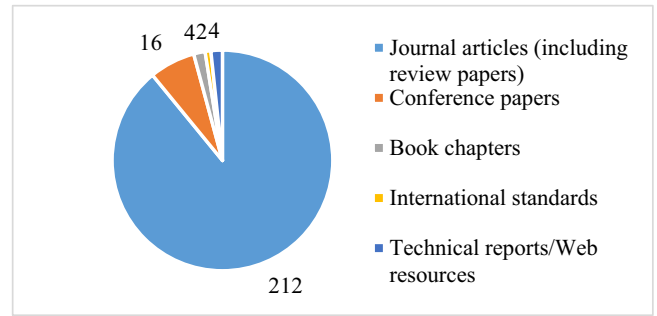


Figure 2. Distribution of the selected documents by publication type.

Table 2

Distribution of the selected literature by research theme.

| Research theme                                                                | Number of studies | Research question | Section |
|-------------------------------------------------------------------------------|-------------------|-------------------|---------|
| Vibration fundamentals and measurement systems                                | 76                | RQ1               | 3       |
| Fault mechanisms and characteristic vibration signatures                      | 103               | RQ2               | 4       |
| Signal processing techniques and intelligent diagnostic methods               | 118               | RQ3               | 5       |
| Standards, practical implementation, research challenges, and emerging trends | 65                | RQ4 and RQ5       | 6       |

The equation of motion for a damped SDOF system subjected to an external excitation force is given by:

$$m\ddot{x} + c\dot{x} + kx = F(t) \quad (1)$$

where  $m$  is the system mass,  $c$  is the damping coefficient,  $k$  is the stiffness,  $x$  is the displacement, and  $F(t)$  is the external excitation force. Although practical rotating machinery usually shows multiple DOF and more complex dynamic behavior, the SDOF model provides a basic foundation for vibration analysis [24].

The main feature of a vibrating system is its natural frequency, the frequency at which the system oscillates after an initial disturbance without ongoing external excitation. When the excitation frequency approaches the natural frequency, resonance occurs. This leads to a large increase in vibration amplitude [25]. Since resonance can accelerate component degradation and increase the risk of failure, its identification and control are crucial in machine design and condition monitoring.

The fundamental principle of the VBCM is that changes in the machine's condition alter the dynamic forces within the system, which in turn affect its vibration response [12]. Mechanical and electromechanical faults often create measurable changes in the vibration amplitude,

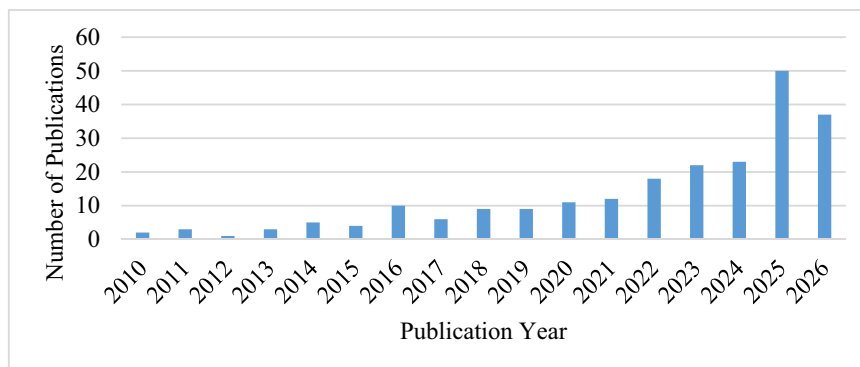


Figure 1. Annual publication trend of the selected studies (2010-2026).

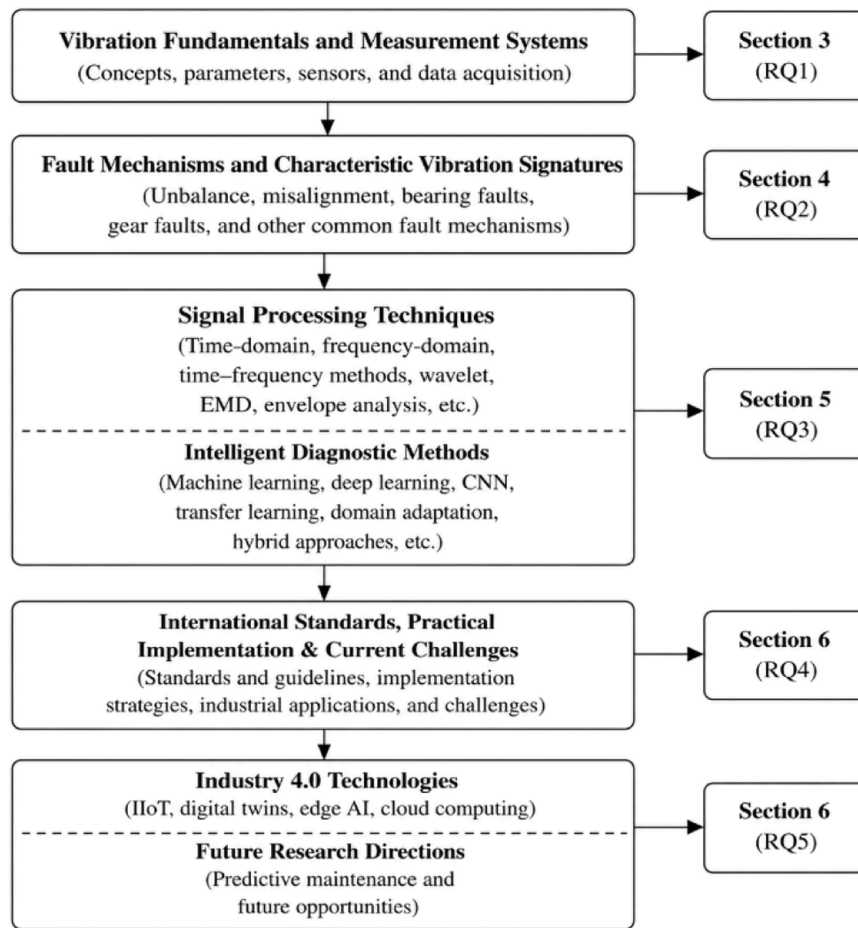


Figure 3. Analytical framework adopted in this review.

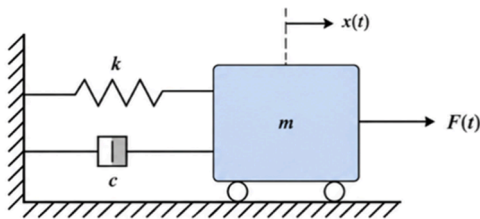


Figure 4. Single-degree-of-freedom (SDOF) vibration model consisting of mass ( $m$ ), stiffness ( $k$ ), damping ( $c$ ), and external excitation force  $F(t)$ .

frequency content, phase relationships, or statistical features before catastrophic failure occurs. By monitoring these changes, vibration analysis helps with fault detection, diagnosis, and maintenance planning.

Because vibration signals contain information from multiple machine components simultaneously, vibration analysis has become one of the most widely used condition-monitoring techniques in rotating machinery [7,8]. Vibration signals can detect a wide range of mechanical, fluid-induced, and electromechanical faults from relatively few measurement locations, making it a cornerstone of predictive and reliability-centered maintenance.

### 3.2. Excitation mechanisms and sources of vibration

Vibration in rotating machinery originates from dynamic excitation forces generated during operation. As shown in Figure 5, these excitation mechanisms arise from mechanical interactions, fluid-flow

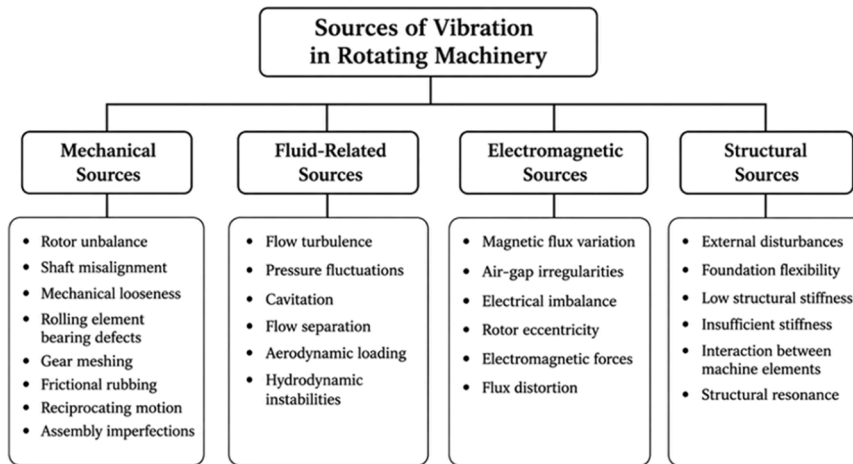
phenomena, electromagnetic effects, structural dynamics, or combinations of these factors [7].

Vibration excitation refers to physical mechanisms that generate dynamic forces within a machine. Vibration propagation describes the transmission of these forces through the machine's structure to measurement locations [26]. During this process, vibration signals can be amplified, attenuated, or distorted depending on structural stiffness, damping, boundary conditions, and resonance effects [27].

Mechanical excitation is the dominant source of vibration in rotating machinery and arises from the interactions among machine components. Common sources are rotor imbalance, shaft misalignment, mechanical looseness, bearing defects, gear meshing, frictional contact, reciprocating motion, and assembly imperfections [28]. These mechanisms generate characteristic frequency components associated with shaft speed, bearing defect frequencies, gear mesh frequencies, and their harmonics, which are widely used as diagnostic indicators.

In fluid-handling machinery, such as pumps, compressors, turbines, and fans, vibrations also arise from fluid dynamic phenomena [29]. Flow turbulence, pressure fluctuations, cavitation, flow separation, aerodynamic loading, and hydrodynamic instabilities generate excitation forces that act on the machine components and surrounding structures. In severe cases, these effects can lead to flow-induced vibration, cavitation damage, blade-pass excitation, or fluid-elastic instability [30].

Electrical machines generate vibration due to electromagnetic forces in motors and generators. Changes in magnetic flux distribution, air-gap irregularities, electrical imbalance, rotor eccentricity, and related electromagnetic interactions create periodic excitation forces that add to machine vibration [31]. These forces interact with mechanical



**Figure 5.** Classification of excitation mechanisms and sources of vibration in rotating machinery.

excitations, further complicating vibration behavior.

Structural excitation arises from the dynamic behavior of machine components, support structures, foundations, and interconnected equipment. External disturbances, foundation flexibility, insufficient stiffness, installation deficiencies, and interactions among machine parts significantly affect vibration transmission and amplification [32].

Resonance occurs when an excitation frequency coincides with a natural frequency of the machine or supporting structure. Under resonant conditions, even small excitation forces can generate disproportionately large vibration amplitudes due to efficient energy transfer into the system. This makes resonance an important factor in machine design, installation, and condition monitoring, as it can amplify other excitation mechanisms and accelerate degradation [33].

In practical industrial machinery, multiple excitation mechanisms often occur simultaneously and interact with one another. The combined effects produce complex vibration responses that change with operating speed, load, environmental conditions, and the machine's health state [18].

### 3.3. Vibration signal characteristics

Vibration signals contain rich information regarding the dynamic behavior and health condition of rotating machinery. The measured vibration response reflects the combined effects of the excitation sources, the machine's structural dynamics, operating conditions, and fault-related phenomena [34]. The analysis of vibration signals can detect, identify, and assess developed faults by analyzing changes in signal characteristics over time. Among various signal attributes, amplitude, frequency, and phase are the main parameters that describe vibration behavior and form the basis of the VBCM.

#### 3.3.1. Amplitude

The amplitude is the magnitude of the vibration and indicates the extent of a component's motion relative to its equilibrium position. It is one of the most common indicators of machine condition because increases in amplitude often reflect deterioration, abnormal operation, or fault development.

Depending on the application, the vibration amplitude can be expressed in terms of displacement, velocity, or acceleration [19]. These parameters are mathematically related through differentiation and integration, but they reveal different frequency ranges and fault sensitivities. Displacement, typically expressed in micrometers ( $\mu m$ ) or millimeters ( $mm$ ), describes physical motion and is particularly useful for low-frequency vibration and shaft motion analysis in large rotating machinery such as turbines, generators, and compressors. Velocity, usually expressed in  $mm/s$ , is widely used as an indicator of vibration

severity as it correlates well with the transmission of mechanical energy through structures. Acceleration, expressed in  $m/s^2$ , is highly sensitive to high-frequency vibrations caused by impacts, friction, and localized defects.

The selection of an appropriate amplitude measure depends on machine type, operating speed, and diagnostic objectives. In practice, acceleration measurements are often preferred because velocity and displacement can be estimated through numerical integration of acceleration signals [18]. However, the accuracy of the derived signals depends on factors such as the accelerometer's low-frequency response, DC bias, measurement noise, temperature drift, filtering, and integration drift. Therefore, dedicated velocity sensors or proximity probes remain preferable when accurate low-frequency velocity or shaft displacement measurements are required, particularly in low-speed machines and large journal-bearing systems. Table 3 presents the main characteristics of commonly used vibration amplitude parameters.

#### 3.3.2. Frequency

Frequency describes the repetition rate of a vibration cycle and is one of the most important diagnostic parameters in vibration analysis. It is usually measured in Hertz (Hz) or cycles per minute for rotating machinery applications [19].

Each component of a rotating machine generates vibration at specific frequencies determined by machine design, operating speed, and dynamic behavior [18,35]. As a result, faults are often linked to unique frequency signatures that can be identified through spectral analysis. Common characteristic frequencies are shaft rotational frequency, bearing defect frequencies, gear mesh frequency, blade-pass frequency, and electrical supply frequency. Figure 6 shows the concept of vibration frequency.

Many faults generate vibrations at characteristic frequencies, so changes in frequency content provide important diagnostic information about the presence, severity, and progression of faults [35]. In practice, vibration signals rarely contain a single frequency component; instead, they consist of multiple interacting frequencies from simultaneous excitation sources [36]. The interpretation of these frequency components is therefore critical for identifying faults in rotating machinery.

**Table 3**  
Comparison of commonly used vibration amplitude parameters.

| Parameter    | Typical unit | Most sensitive to          |
|--------------|--------------|----------------------------|
| Displacement | $\mu m, mm$  | Low-frequency vibration    |
| Velocity     | $mm/s$       | Medium-frequency vibration |
| Acceleration | $m/s^2$      | High-frequency vibration   |

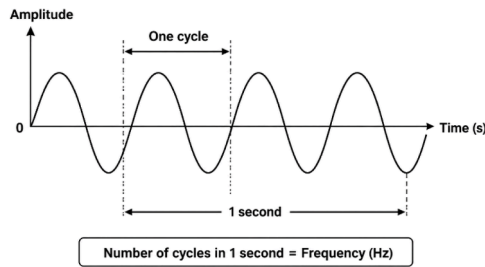


Figure 6. Representation of vibration frequency.

### 3.3.3. Phase

Phase describes the relative timing between vibration signals and is expressed in degrees ( $0^\circ$  to  $360^\circ$ ) [18]. Unlike amplitude and frequency, which describe vibration magnitude and repetition rate, phase provides information about the spatial and temporal relationship between vibration sources and measurement locations.

Phase measurements can be relative or absolute [37]. Relative phase compares two vibration signals measured at different locations and indicates whether their motions happen at the same time or with a delay. This measure is particularly useful for identifying vibration transmission paths and dynamic relationships within machinery. The concept of relative phase difference is shown in Figure 7.

Absolute phase is obtained using a rotational reference signal such as a tachometer or keyphasor [38]. This method establishes a direct relationship between the measured vibration response and the shaft's angular position. It is widely used in balancing, alignment verification, and rotor dynamic studies. Figure 8 shows the concept of absolute phase measurement referenced to shaft rotational position.

Phase analysis provides important diagnostic information not available from amplitude and frequency alone, particularly when different faults exhibit similar amplitude and frequency characteristics [39].

## 3.4. Vibration measurement systems

Vibration measurement systems generally consist of sensors, mounting hardware, data acquisition equipment, and analysis software. The effectiveness of these systems depends on the proper selection, installation, and integration of all measurement components.

### 3.4.1. Sensor technologies

Various sensor technologies are used for measuring vibration in rotating machinery. Sensor selection depends on factors such as machine type, operating environment, frequency range, installation constraints, and diagnostic objectives [40]. Vibration sensors fall into two main classes: contact and non-contact sensors. Contact sensors are mounted directly on machine structures, whereas non-contact sensors measure vibration without physical contact. Common vibration-sensing technologies are proximity probes, velocity sensors, accelerometers, laser vibrometers, and millimeter-wave (mmWave) sensing systems [20].

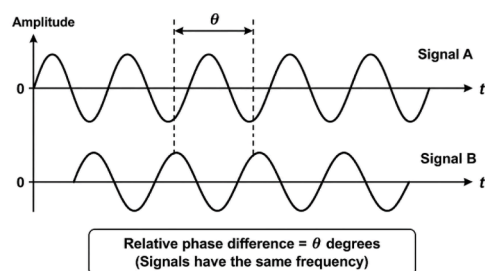


Figure 7. Relative phase difference between two vibration signals with identical amplitude and frequency but different phase angles.

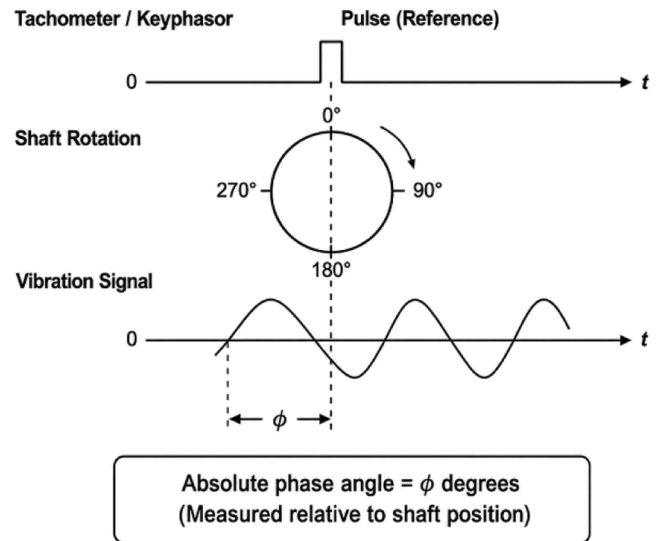


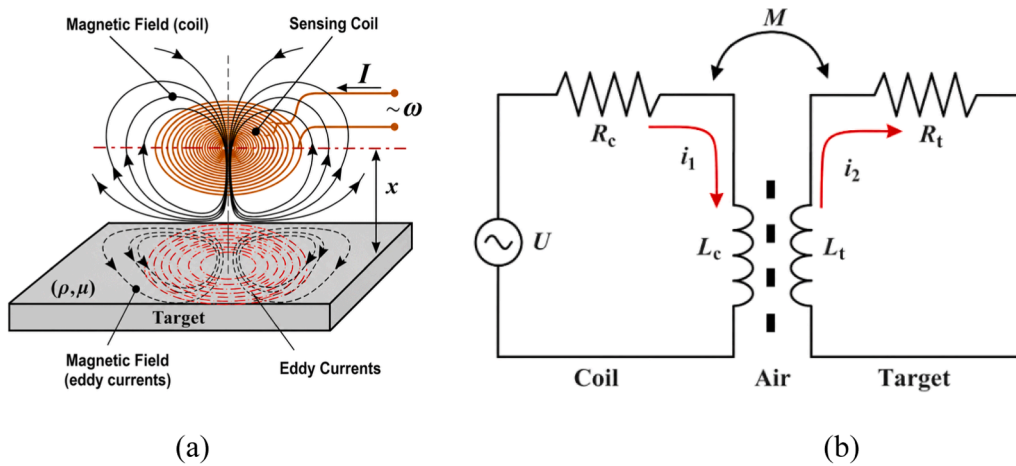
Figure 8. Absolute phase measurement referenced to the shaft rotational position using a rotational reference signal.

Among these, proximity probes are non-contact displacement sensors that measure the relative displacement of a rotating shaft with respect to the bearing housing using eddy-current principles [41,42]. As shown in Figure 9, variations in the gap between the probe tip and the conductive shaft cause corresponding changes in output voltage, enabling direct measurement of shaft motion. These sensors are particularly suitable for large rotating machinery such as turbines, compressors, and generators, where shaft motion is not transmitted to the casing [43]. As a result, these sensors are commonly used for rotor dynamic analysis, orbit plotting, clearance monitoring, and machine protection systems.

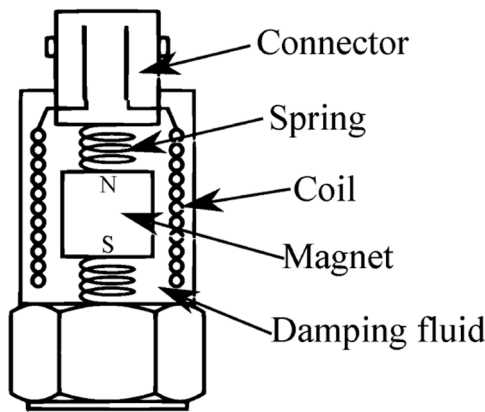
Velocity sensors, or seismic transducers, generate an electrical signal proportional to vibration velocity via electromagnetic induction. Mounted on machine housings, they provide a direct measure of overall vibration severity and are commonly used in machinery condition monitoring and vibration assessment applications [44]. As shown in Figure 10, a velocity sensor is based on a seismic mass-spring-damper system integrated with a coil-magnet arrangement, which converts mechanical motion into an electrical output proportional to velocity. However, these sensors have a narrower frequency response than accelerometers, which limits their sensitivity to high-frequency fault signals.

Accelerometers are the most widely used vibration sensors in modern condition-monitoring systems due to their broad frequency bandwidth, high sensitivity, compact size, and versatility [19]. Most industrial accelerometers use piezoelectric elements, which generate an electrical charge proportional to the applied acceleration [43]. These sensors typically employ a seismic mass-spring-damper configuration, as shown in Figure 11. Relative motion between the mass and housing creates stress in the piezoelectric crystal, producing an electrical output signal. Acceleration signals can be integrated to obtain velocity and displacement, enabling a single sensor to support multiple diagnostic applications.

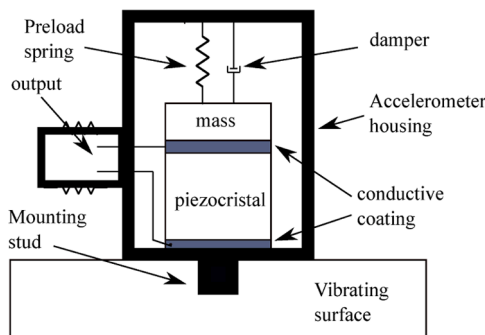
Non-contact vibration measurement technologies are gaining more attention because they eliminate the need for physical sensor attachment and enable measurements in inaccessible or harsh environments [19]. Examples of non-contact sensors are laser Doppler vibrometers and mmWave sensing systems. Laser vibrometers measure vibration velocity using the Doppler shift of reflected laser light and provide high measurement accuracy over a wide frequency range [43]. In contrast, mmWave systems estimate vibration motion from reflected electromagnetic waves and are particularly suited for harsh industrial



**Figure 9.** Operating principle and equivalent representation of an eddy-current proximity probe: (a) magnetic field distribution and eddy-current generation; (b) equivalent electrical model [41].



**Figure 10.** Schematics of a velocity sensor and its main components [7].



**Figure 11.** Internal structure of a piezoelectric accelerometer [45].

environments or locations where installing sensors directly is impractical [46].

### 3.4.2. Sensor mounting considerations

The quality of vibration measurements depends strongly on proper installation and mounting practices. Even high-performance sensors can yield inaccurate results if improperly mounted, leading to the loss of important frequency components, distortion of vibration amplitudes, and reduced diagnostic reliability [47]. Thus, sensor mounting directly influences measurement accuracy, repeatability, and long-term consistency.

Proper sensor mounting ensures efficient transfer of vibration energy

from the machine structure to the sensing element. This requires a rigid mechanical coupling between the sensor and its mounting surface while reducing the effects of structural flexibility, surface contamination, mounting resonance, and external disturbances [48]. Improper mounting can reduce high-frequency response, introduce amplitude errors, and lead to misleading diagnostic interpretations [14].

Sensor placement is equally important for obtaining meaningful vibration data [49]. Sensors should be installed at locations where vibration transmission is most effective and fault-related signatures are most pronounced. In rotating machinery, preferred measurement locations are typically near bearings, gearboxes, and coupling regions, as these components provide direct transmission paths for vibration generated by mechanical defects and dynamic interactions [50]. Bearing housings are generally preferred because they facilitate efficient vibration transmission from rotating and stationary machine elements with minimal attenuation or distortion. On the other hand, sensors should not be mounted on flexible or structurally weak parts, such as thin covers, guards, piping, or access panels. These locations can create local resonance effects and structural filtering that distort the signals captured [51].

The direction of measurement is another important consideration. Vibration is commonly measured in the horizontal, vertical, and axial directions because different fault mechanisms create unique directional vibration signatures [52]. For example, rotor imbalance predominantly generates radial vibrations, whereas shaft misalignment often results in significant axial vibration components [53].

Finally, consistent sensor positioning is essential for condition monitoring programs based on trend analysis. Taking repeated measurements at the same locations, orientations, and operating conditions improves data repeatability and enables a reliable long-term assessment of machine conditions [9]. Because of this, standardized measurement points are mostly used in predictive maintenance programs.

### 3.4.3. Data acquisition and signal conditioning

Most vibration sensors produce analog electrical outputs that require conditioning before digital acquisition. Signal conditioning comprises processes that ensure signal quality and compatibility with data acquisition systems, such as amplification, filtering, impedance matching, charge conversion, and noise reduction [54]. Amplification adjusts low-level sensor outputs to suitable voltage ranges for analog-to-digital conversion. It is particularly important for piezoelectric accelerometers, where charge signals are converted into voltage using charge amplifiers. Modern conditioning units also provide sensor excitation, calibration support, and protection against electrical noise and interference [19].

After amplification, filtering is applied to improve signal integrity

and remove unwanted components. High-pass filters eliminate low-frequency drift and DC offsets. Low-pass filters suppress high-frequency noise [55]. In some applications, band-pass filtering isolates frequency bands associated with specific fault mechanisms, thereby improving diagnostic features [56].

Following signal conditioning, analog signals are converted into digital form using analog-to-digital converters (ADCs). The sampling frequency is a critical parameter, as insufficient sampling can lead to loss of diagnostic information [57]. According to the Nyquist theorem, the sampling rate must be at least twice the highest frequency component in the signal [58]. However, practical systems typically use higher sampling rates to improve frequency resolution and ensure reliable fault detection. To maintain signal quality, aliasing must be controlled. This occurs when frequency components above the Nyquist limit are mapped into lower frequency ranges, creating misleading spectral content [59]. Therefore, anti-aliasing filters are applied before digitization to remove unsampled high-frequency components [60].

Modern vibration monitoring systems often use multi-channel data acquisition to collect signals from multiple sensor locations simultaneously [61]. This capability is essential for evaluating phase relationships, vibration transmission paths, and dynamic interactions among machine components. Diagnostic performance improves with synchronized acquisition, which ensures all channels share a common time reference [62,63]. Advances in acquisition hardware, embedded processing, and industrial communication technologies allow real-time processing, remote monitoring, automated fault detection, and integration with predictive maintenance platforms, supporting more intelligent machine health management systems [19].

#### 3.4.4. Monitoring strategies

The effectiveness of the VBCM depends not only on the measurement quality but also on the strategy used to collect and analyze the vibration data. Monitoring strategies define how vibration information is collected, processed, and used for fault detection and condition monitoring in rotating machinery. These approaches are generally classified into two categories: permanent (online) and intermittent (offline or route-based) monitoring [64].

Permanent monitoring uses permanently installed sensors connected to dedicated acquisition and analysis units that continuously collect vibration data during machine operation. This method is particularly effective for detecting transient events and rapidly developing faults that are not captured during periodic inspections. It is commonly applied to critical machinery, such as turbines, compressors, large pumps, and generators, where unexpected failures can result in significant economic losses, safety hazards, and production downtime [65]. In modern industrial facilities, permanent monitoring systems are often integrated with plant-wide platforms, such as SCADA, DCS, and CMMS, to support automated decision-making and predictive maintenance [66]. However, implementation requires substantial investments in sensors, cabling, conditioning hardware, communication infrastructure, and software.

Intermittent vibration monitoring, also known as route-based or periodic monitoring, involves measurements acquired at scheduled intervals using portable data collectors, hand-held instruments, or wireless devices [65]. This approach balances diagnostic capability and implementation effort cost-effectively, making it suitable for large equipment populations. For reliable fault detection, inspection intervals must be carefully selected to detect fault progression before failure occurs. Intermittent monitoring is particularly attractive because of its flexibility, scalability, and ease of implementation [7]. In industrial practice, hybrid monitoring strategies are increasingly adopted, combining permanent monitoring for critical assets with intermittent monitoring for less critical equipment [65].

## 4. Fault mechanisms in rotating machinery and characteristic vibration signatures

This section presents the fault mechanisms in rotating machinery and their characteristic vibration signatures, addressing RQ2.

### 4.1. Mechanical faults

Mechanical faults are the most common causes of abnormal vibration in rotating machinery. The most common mechanical faults are shaft unbalance, shaft misalignment, bent shafts, and looseness [7].

#### 4.1.1. Shaft unbalance

Shaft unbalance is one of the most common faults in rotating machinery. It occurs when the center of mass of a rotating component does not coincide with its geometric axis of rotation, resulting in a mass eccentricity that generates centrifugal forces during operation [67]. The resulting unbalanced force is defined as [68]:

$$F = m \cdot e \cdot \omega^2 \quad (2)$$

where  $F$  is the unbalance force (N),  $m$  is the unbalance mass (kg),  $e$  is the eccentricity (m), and  $\omega$  is the angular speed (rad/s), given by  $\omega = \frac{2\pi N}{60}$ . The physical mechanism of shaft unbalance, and the parameters associated with the unbalance force are shown in Figure 12.

Several forms of imbalance occur in practice, including static and dynamic imbalances [69]. Static unbalance arises from a single eccentric mass distribution, whereas dynamic unbalance represents the most general case, involving a combination of force and moment effects distributed along the rotor length. In industrial machinery, mixed unbalance conditions are common and typically require multi-plane balancing corrections [70]. Unbalance results from manufacturing tolerances, assembly errors, uneven material distribution, erosion, corrosion, mechanical wear, contamination buildup, or the loss of balancing weights. In equipment such as fans, blowers, pumps, and compressors, the gradual accumulation of dirt or process material is a common cause of increasing mass eccentricity during operation.

An unbalanced rotor generates a rotating excitation force synchronized with shaft motion, producing a periodic vibration response at the same rotational frequency. The most characteristic feature of shaft unbalance is a dominant vibration component at the fundamental rotational frequency, commonly referred to as the  $1\times$  running speed component [71]. The excitation force generated by rotor unbalance increases approximately with the square of rotational speed, as indicated by Equation (2) [72,73]. However, the measured vibration amplitude depends not only on the excitation force but also on the dynamic characteristics of the rotor-bearing-support system, including its stiffness, damping, frequency-dependent dynamic compliance, support conditions, and proximity to critical speeds. Consequently, the measured vibration response does not always increase proportionally with the square of rotational speed.

The resulting vibration is predominantly radial because the centrifugal force acts perpendicular to the shaft centerline. Phase information

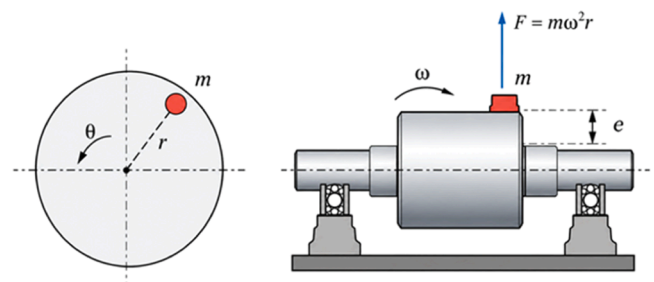


Figure 12. Shaft unbalance.

provides an important diagnostic indicator for unbalance detection. Pure unbalance typically exhibits a stable phase relationship relative to shaft rotation, and vibration measurements at opposite sides of the rotor often show phase differences approaching  $180^\circ$ . In contrast, measurements in orthogonal directions (horizontal and vertical) exhibit phase shifts of approximately  $90^\circ$ , consistent with circular vibration motion [74].

From a diagnostic perspective, shaft unbalance is characterized by a dominant  $1\times$  running speed frequency component, high radial vibration amplitude, low harmonic content, and stable phase behavior [71,75]. If uncorrected, imbalance increases bearing loads, shaft stresses, seal degradation, and fatigue damage.

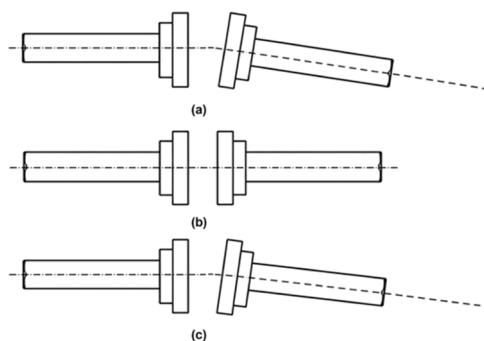
#### 4.1.2. Shaft misalignment

Shaft misalignment occurs when the rotational centerlines of coupled shafts are not collinear during operation [76,77]. It is among the most frequently encountered installation- and maintenance-related faults and a major contributor to elevated vibration levels, excessive mechanical loading, and reduced machinery reliability. Misalignment results from improper installation, thermal expansion, foundation settlement, pipe strain, structural distortion, coupling wear, or maintenance errors. Thermal growth is particularly important in high-temperature machinery because alignment conditions change substantially between cold and normal operating states [78].

Misalignment is generally classified into angular, parallel (offset), and combined misalignment [79]. Figure 13 shows typical angular and parallel shaft misalignment configurations. Angular misalignment occurs when shaft centerlines intersect but are not parallel, whereas parallel misalignment exists when shaft centerlines remain parallel but are laterally displaced. In industrial applications, combined shaft misalignment is the most common condition because both angular and offset errors frequently occur simultaneously during installation or operation.

Unlike rotor unbalance, which generates rotating radial forces, shaft misalignment introduces cyclic forces and moments through the coupling and shaft system. These forces vary throughout each shaft revolution and are transmitted to bearings, seals, couplings, and support structures, resulting in increased vibration and elevated mechanical stresses. The vibration response associated with misalignment is generally more complex than that of unbalance. The vibration spectrum commonly contains significant components at the  $1\times$  running speed frequency and elevated vibration at the  $2\times$  running speed frequency, particularly in cases of angular misalignment [80]. As fault severity increases, additional higher-order harmonics ( $3\times$ ,  $4\times$ , and higher) appear [81]. Elevated axial vibration is another important diagnostic characteristic because coupling forces continuously act to accommodate angular and offset deviations between connected shafts [82]. Machines with significant misalignment also commonly exhibit abnormal phase relationships across the coupling, particularly in the axial direction.

Shaft misalignment is commonly diagnosed using vibration spectrum



**Figure 13.** Types of shaft misalignment: (a) angular misalignment, (b) parallel (offset) misalignment, and (c) combined misalignment.

analysis by identifying the characteristic  $1\times$  and  $2\times$  running-speed components, together with higher-order harmonics in severe cases. Phase analysis across the coupling, particularly in the axial direction, is widely used to distinguish misalignment from rotor unbalance. Elevated axial vibration also serves as an important diagnostic indicator [80–82]. If not corrected, shaft misalignment can lead to accelerated bearing degradation, coupling damage, shaft fatigue, seal wear, excessive power consumption, and premature machine failure [78].

#### 4.1.3. Bent shaft

A bent shaft fault occurs when the shaft centerline deviates from its intended geometric axis because of permanent deformation or temporary thermal distortion. Shaft bending can result from manufacturing defects, improper handling, excessive mechanical loading, rotor-rub events, residual stresses, or uneven thermal expansion during operation [83]. The physical mechanism of shaft bending is shown in Figure 14, where  $\delta$  is the shaft deflection from the ideal centerline.

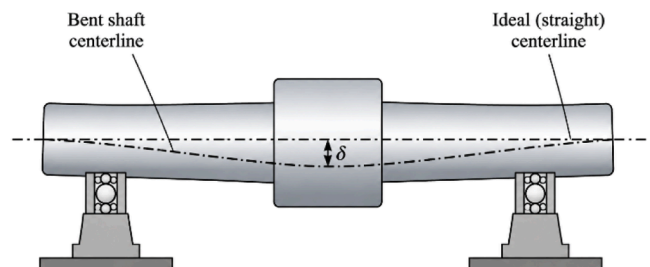
The presence of shaft curvature causes the rotor to rotate eccentrically, generating periodic excitation forces that alter the dynamic behavior of the rotor-bearing system [84]. Thermally induced shaft bending is of particular importance in large rotating machines such as turbines and generators. Uneven temperature distributions during startup, shutdown, or transient operating conditions cause temporary shaft distortion, producing vibration characteristics that change with operating temperature and machine load [85].

Because its vibration characteristics overlap with those of rotor unbalance and shaft misalignment, bent shaft faults are often difficult to diagnose using vibration measurements alone. The vibration response is commonly dominated by a significant  $1\times$  running speed component [86]. Elevated axial vibration levels are frequently observed alongside radial vibration because shaft deformation alters the geometric relationship between the rotating and stationary components. Furthermore, phase behavior is often more complex or unstable than that associated with pure rotor unbalance, reflecting the distributed nature of shaft deformation. Temperature-dependent changes in vibration behavior frequently provide important evidence for distinguishing bent shafts from other mechanically induced vibration sources.

Reliable diagnosis of a bent shaft typically requires the combined interpretation of vibration spectra, phase measurements, shaft orbit plots, and operational observations [87]. The fault is generally identified by the presence of a dominant  $1\times$  running speed component, elevated axial vibration, complex or unstable phase behavior, and sensitivity to operating temperature and load conditions [86]. Because these characteristics overlap with those of rotor unbalance and shaft misalignment, accurate diagnosis often requires integrating multiple vibration analysis techniques with machine operating data.

#### 4.1.4. Mechanical looseness

Mechanical looseness is characterized by excessive clearance, insufficient rigidity, or loss of structural integrity between components that are intended to remain securely connected during operation. This is particularly critical because it introduces strong nonlinear behavior into the system dynamics [88]. Unlike faults such as imbalance or



**Figure 14.** Bent shaft.

misalignment, looseness is characterized by intermittent contact, variable stiffness, and repeated impact phenomena, resulting in highly irregular vibration responses.

Mechanical looseness is commonly classified into rotational, bearing, and structural looseness [89]. Rotational looseness involves excessive clearance between rotating and stationary components; bearing looseness results from improper fits between the shaft, bearing, and housing; and structural looseness arises from insufficient rigidity in supports, foundations, or machine attachments.

In the frequency domain, mechanical looseness is typically identified by multiple running-speed harmonics ( $1\times$ ,  $2\times$ ,  $3\times$ , and higher). In severe cases, subharmonic components, such as  $0.5\times$  running speed, may also appear owing to nonlinear impact mechanisms [90]. In the time domain, the vibration response is strongly non-sinusoidal and often exhibits impact transients, amplitude modulation, irregular peaks, and unstable vibration amplitudes [91]. These nonlinear characteristics make mechanical looseness one of the most complex vibration signatures in rotating machinery [39].

Mechanical looseness is commonly diagnosed using time-domain analysis to identify impact transients and waveform irregularities, together with frequency-spectrum analysis to detect multiple harmonics and subharmonic components associated with nonlinear impacts. Because looseness frequently develops secondary to faults such as rotor unbalance, shaft misalignment, or bearing degradation, reliable diagnosis often requires distinguishing its nonlinear vibration characteristics from those of coexisting faults [39,90,91]. If not corrected, mechanical looseness can significantly amplify vibration, leading to progressive wear, fastener loosening, foundation damage, fatigue failure, and catastrophic machine failure [91].

#### 4.1.5. Additional mechanical faults

In addition to the faults discussed above, rotating machinery can experience other mechanical defects, including rotor rub, cracked shafts, soft foot, coupling defects, resonance-related problems, and structural weaknesses [92]. These faults arise from different physical mechanisms, such as frictional contact, material cracking, improper machine installation, coupling deterioration, structural flexibility, or resonance excitation, and may significantly influence machine dynamic behavior.

Although these faults are generally less common or more application-specific, they can also generate distinctive vibration signatures useful for diagnosis. Depending on the fault mechanism, the vibration response includes harmonic or subharmonic components, impact transients, nonlinear responses, amplitude modulation, or resonance-induced vibration amplification. In practice, multiple mechanical faults often coexist and interact, and yield complex vibration patterns that require careful interpretation [93].

Reliable diagnosis of these additional mechanical faults generally requires the combined interpretation of time-domain, frequency-domain, and phase analyses, together with knowledge of the machine operating conditions and mechanical configuration. Because multiple faults may occur simultaneously, integrated vibration analysis is often necessary to distinguish interacting fault mechanisms and improve diagnostic reliability [93].

### 4.2. Fluid-film bearing instabilities

Fluid-film bearings are widely used in high-speed rotating machinery, such as steam turbines, gas turbines, centrifugal compressors, engines, and large generators, because they support heavy loads while enabling smooth rotor operation [29]. Unlike rolling-element bearings, fluid-film bearings rely on a thin lubricant film to separate the rotating shaft from the bearing surface, thereby preventing direct metal-to-metal contact. However, fluid-film bearings are susceptible to self-excited dynamic instabilities that can significantly affect machine reliability and operational safety [94,95]. Among the most important and widely studied instabilities are oil whirl and oil whip, which arise from dynamic

interactions among the rotating shaft, the lubricant film, and the bearing structure [96].

#### 4.2.1. Oil whirl

Oil whirl is a self-excited vibration that arises when hydrodynamic forces in the lubricant film become dynamically unstable [97]. During normal operation, the rotating shaft drags the lubricant around the bearing, generating a pressure field that supports the rotor load. Under certain operating conditions, destabilizing fluid-film forces exceed the stabilizing effects of rotor stiffness and damping, causing the shaft centerline to precess within the bearing clearance [94]. The onset of oil whirl is influenced by bearing geometry, lubricant viscosity, rotor mass, bearing load, operating speed, shaft flexibility, bearing clearance, and the bearing length-to-diameter ( $L/D$ ) ratio [95,97–99].

Oil whirl produces a characteristic subsynchronous vibration component, typically occurring at approximately  $0.4\times$ – $0.5\times$  the running speed [98]. The exact frequency depends on the bearing design and operating conditions [99]. In some cases, oil whirl may be masked by rotor unbalance in shaft response measurements but remains detectable in bearing vibration signals [100]. The vibration is also characterized by elevated radial vibration and enlarged circular or elliptical shaft orbits associated with forward rotor precession. The absence of backward precession is an important feature distinguishing oil whirl from other instability mechanisms, such as rub-induced vibration [94,97,98].

Oil whirl is commonly diagnosed through vibration spectrum analysis by identifying the characteristic  $0.4\times$ – $0.5\times$  subsynchronous frequency component, together with orbit analysis to observe forward shaft precession within the bearing clearance. Bearing vibration measurements are particularly useful when shaft-response measurements are influenced by rotor unbalance [94,97,98,100]. Continued increases in rotational speed may cause oil whirl to transition into the more severe oil-whip instability.

#### 4.2.2. Oil whip

Oil whip is a more severe form of fluid-film instability that develops when the subsynchronous vibration associated with oil whirl excites a lateral natural frequency of the rotor-bearing system [97,101]. Once resonance is reached, vibration amplitudes increase sharply, producing a sustained self-excited instability [95]. The susceptibility to oil whip depends on rotor dynamics, bearing design, lubricant properties, operating conditions, and bearing wear [94,95,102].

Unlike oil whirl, whose frequency varies with shaft speed, oil whip is characterized by subsynchronous vibration locked to a rotor natural frequency, remaining nearly constant even as the shaft speed continues to increase [101]. It is also associated with large orbit amplitudes, sustained high vibration levels, and resonance-induced rotor motion. If left unchecked, oil whip can result in rotor-to-stator contact, bearing damage, excessive heat generation, structural fatigue, and serious risks to machine integrity [94,95,97].

Oil whip is commonly diagnosed by identifying a persistent subsynchronous frequency locked to the rotor natural frequency, together with large orbit amplitudes and sustained high vibration levels over a wide operating-speed range [97]. Mitigation typically involves modifications to bearing design, lubrication conditions, rotor dynamic stiffness, or operating parameters. Advanced bearing configurations have also been shown to suppress oil whirl and oil whip while extending the stable operating range [94,95,99].

#### 4.2.3. Whirl/whip synchronization

In rotor systems supported by full-floating ring bearings, an additional instability may arise from interactions between the inner and outer lubricant films [96]. As rotational speed increases, the instability frequencies associated with the two lubricant films may become synchronized and eventually lock onto a rotor natural frequency, producing a condition commonly referred to as “total instability” [97].

Whirl/whip synchronization is characterized by strong subsynchronous vibration, large shaft-orbit amplitudes, and a rapid increase in vibration amplitude as the operating condition approaches the instability threshold [96,97].

Diagnosis is primarily based on identifying strong subsynchronous vibration together with orbit analysis and operating-speed trends that indicate synchronization of the lubricant-film instabilities. Although less common than conventional oil whirl and oil whip, early detection is essential because this instability can rapidly lead to severe rotor damage in high-speed machines equipped with full-floating ring bearings [96, 97].

#### 4.3. Rolling element bearing faults

Rolling element bearings are critical components in rotating machinery, supporting radial and axial loads, maintaining precise shaft alignment, and enabling low-friction rotation. Because of continuous mechanical loading and contact stresses, bearings are among the most failure-prone components in industrial systems. Statistics indicate that bearing failures account for approximately 40% of motor failures [103]. In the wind energy sector, bearings contribute to nearly 70% of gearbox failures and 21-70% of generator failures, depending on operating conditions and machine configuration [104]. These statistics underscore the importance of effective bearing condition monitoring and early fault detection.

##### 4.3.1. Bearing fault mechanisms

A rolling element bearing consists of four principal components: the inner race, outer race, rolling elements, and cage (retainer), as shown in Figure 15. Defects can develop in any of these components due to rolling contact fatigue, inadequate lubrication, contamination, excessive loading, improper installation, electrical erosion, corrosion, or manufacturing imperfections [105].

The most common bearing faults involve localized defects in the outer race, inner race, rolling elements, or cage. These defects create surface discontinuities that disrupt normal rolling contact. As rolling elements repeatedly pass over damaged regions, impulsive excitation forces are generated and transmitted through the bearing and surrounding structure [106]. These impacts occur at characteristic intervals determined by bearing geometry and shaft rotational speed, producing distinct vibration signatures that form the basis of vibration-based bearing fault diagnosis.

During early degradation, defects are typically microscopic and produce weak vibration responses that are often difficult to distinguish from background noise. As the damage progresses, the impact magnitude increases, resulting in higher vibration amplitudes, broader spectral content, and more pronounced fault-related frequency components [107]. Continued deterioration can lead to surface spalling, increased

friction, excessive heat generation, lubricant degradation, and ultimately catastrophic bearing failure [108].

##### 4.3.2. Characteristic bearing fault frequencies

One of the most important features of rolling element bearing diagnostics is the presence of characteristic defect frequencies directly related to bearing geometry and rotational speed [109]. These frequencies arise from kinematic interactions among the rolling elements, races, and cage. Because each damaged component generates a distinct characteristic vibration frequency, these frequencies provide reliable indicators for fault localization and condition monitoring. The principal characteristic frequencies are the Ball Pass Frequency of the Outer Race (BPFO), Ball Pass Frequency of the Inner Race (BPFI), Ball Spin Frequency (BSF), and Fundamental Train Frequency (FTF). For a bearing with a stationary outer race, these frequencies are given by [105,110]:

$$BPFO = \frac{nf_r}{2} \left( 1 - \frac{d}{D} \cos\theta \right) \quad (3)$$

$$BPFI = \frac{nf_r}{2} \left( 1 + \frac{d}{D} \cos\theta \right) \quad (4)$$

$$FTF = \frac{f_r}{2} \left( 1 - \frac{d}{D} \cos\theta \right) \quad (5)$$

$$BSF = \frac{D}{2d} \left( 1 - \left( \frac{d}{D} \cos\theta \right)^2 \right) \quad (6)$$

where  $f_r$  is the shaft rotational frequency,  $n$  is the number of rolling elements,  $d$  is the rolling-element diameter,  $D$  is the pitch diameter of the bearing, and  $\theta$  is the contact angle between the rolling element and the raceway.

Each characteristic frequency is associated with a specific defect location: BPFO corresponds to outer-race defects, BPFI to inner-race defects, BSF to rolling-element defects, and FTF to cage or retainer faults [111]. These frequencies serve as fundamental diagnostic indicators for fault identification and localization, as provided in Table 4.

In practical applications, defect-related frequency components often appear alongside harmonics and sidebands spaced at the shaft rotational frequency [112,113]. These modulation effects arise from load variations, slip, and time-varying contact conditions, as shown in Figure 16. The resulting sideband patterns provide additional diagnostic information regarding fault severity and progression.

##### 4.3.3. Characteristic vibration signatures and diagnostic methods

The vibration response of rolling element bearing defects is typically impulsive and non-stationary rather than purely harmonic. Localized damage generates short-duration impact forces when rolling elements interact with defect surfaces. These impacts excite the natural resonant frequencies of the bearing and surrounding machine structure, resulting in high-frequency vibration bursts [114].

Bearing faults, therefore, typically appear as broadband high-frequency vibration energy modulated by lower-frequency mechanical components [115]. In the time domain, this behavior appears as repetitive impulses whose repetition rate corresponds to the characteristic defect frequency. However, during the early stages of fault development,

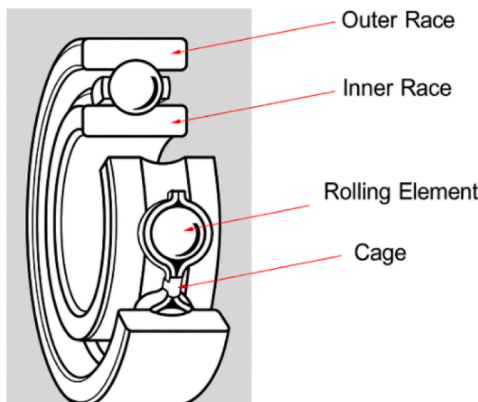
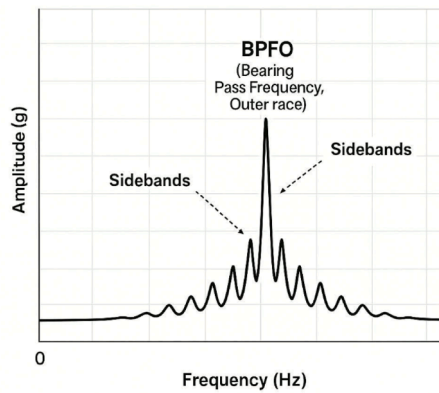


Figure 15. Elements of the rolling element bearing.

Table 4

Characteristic rolling element bearing fault frequencies and associated defect locations.

| Frequency | Associated component / fault location |
|-----------|---------------------------------------|
| BPFO      | Outer-race defects                    |
| BPFI      | Inner-race defects                    |
| BSF       | Rolling-element defects               |
| FTF       | Cage / retainer defects               |



**Figure 16.** Simplified frequency-domain representation of bearing fault signatures.

these impulses are often weak and masked by background vibration and operational noise, making direct detection difficult [116].

As defect severity increases, impact energy rises, leading to higher vibration amplitudes, spectral broadening, and clearer fault-related frequency components and harmonics. In advanced stages of degradation, multiple defects may coexist, producing complex vibration patterns characterized by dense spectral content, elevated noise floors, and strong modulation effects [117].

These characteristic vibration signatures form the basis for rolling element bearing fault diagnosis and determine the selection of appropriate signal-processing techniques. Because bearing fault signals are highly impulsive and non-stationary, conventional indicators such as RMS are often insufficient for early fault detection. Therefore, advanced signal-processing techniques, such as envelope analysis, spectral kurtosis, and resonance demodulation, are commonly employed to extract hidden fault signatures, improve diagnostic sensitivity, and facilitate the identification and localization of bearing defects [118,119].

#### 4.4. Gear faults

##### 4.4.1. Gear fault mechanisms

Gear transmission systems are widely used in industrial machinery to transmit power and modify speed or torque between rotating shafts. Because gears operate under repeated cyclic contact loading, they are inherently susceptible to wear and damage mechanisms that can significantly degrade machine performance, transmission efficiency, and operational reliability [120]. Gear defects alter the dynamic interaction between meshing teeth, introducing additional excitation forces into the drivetrain and modifying the vibration response. The resulting vibration signatures depend on gear geometry, rotational speed, transmitted load, lubrication conditions, and fault severity [121].

Gear faults develop gradually through progressive wear or occur suddenly because of excessive loading, impact events, or severe mechanical damage. The most common failure mechanisms are tooth wear, pitting, spalling, scuffing, cracking, chipped teeth, broken teeth, gear eccentricity, and manufacturing defects [122,123]. Tooth wear involves the progressive removal of material from the tooth surface due to repeated frictional contact during meshing, whereas pitting and spalling mainly result from rolling-contact fatigue, causing surface or subsurface crack initiation and eventual material detachment [124,125]. Cracks commonly originate in high-stress regions, such as the tooth root, and propagate under cyclic loading, leading to tooth fracture [126]. Gear eccentricity and manufacturing defects further degrade meshing accuracy by introducing periodic variations in tooth engagement and load distribution [127].

The initiation and progression of these failure mechanisms are strongly influenced by operating and environmental conditions. Lubrication deficiencies, contamination, overload, improper alignment,

installation errors, and manufacturing imperfections increase contact stresses and accelerate deterioration [128]. As defects develop, tooth contact becomes increasingly irregular, reducing smooth load transmission and generating additional dynamic excitation forces within the gearbox. These changes form the physical basis for vibration-based gear fault diagnosis.

##### 4.4.2. Characteristic gear fault frequencies

Under normal operating conditions, gearbox vibration is dominated by the gear mesh frequency (GMF), which is the rate at which the gear teeth engage. The gear mesh frequency is expressed as [129]:

$$GMF = N \times Z \quad (7)$$

where  $N$  is the shaft rotational frequency and  $Z$  is the number of teeth on the gear.

Because each tooth engagement generates a periodic excitation force, vibration energy is naturally concentrated at the gear mesh frequency and its harmonics, even in healthy gear systems [130]. These components constitute the primary spectral features of a gearbox and provide a baseline for identifying fault-related changes.

As gear defects develop, the meshing process is disturbed by localized variations in tooth geometry, stiffness, and load distribution, altering force transmission between mating gears [131]. Consequently, vibration amplitudes at the GMF and its harmonics increase with fault severity. Localized defects also produce periodic fluctuations in transmitted load, giving rise to amplitude modulation and, in some cases, frequency modulation of the gear mesh vibration signal [132].

The modulation process gives rise to characteristic sideband structures surrounding the gear mesh frequency and its harmonics. These sidebands are typically spaced at the shaft rotational frequency because the damaged tooth re-enters the meshing zone once per shaft revolution [133]. As damage progresses, both GMF amplitudes and sideband components increase, making these spectral features highly useful for fault detection, localization, and severity assessment [134].

##### 4.4.3. Characteristic vibration signatures and diagnostic methods

The vibration response of a gearbox depends on the defect type, location, and severity. Localized faults such as cracked, chipped, or broken teeth generate transient impact forces whenever the damaged tooth enters the meshing zone. These repeated impacts produce impulsive excitation synchronized with tooth engagement, and their intensity generally increases as damage propagates [135].

In the time domain, gear defects are typically characterized by impulsive waveforms, periodic amplitude fluctuations, and increasingly non-stationary behavior due to repeated tooth impacts. In the frequency domain, fault development is reflected by increased vibration amplitudes at the GMF, elevated harmonic content, and progressively stronger sidebands around the GMF and its harmonics [136]. These characteristic vibration signatures provide important information for fault detection, localization, severity assessment, and progression monitoring.

As damage becomes more severe, the vibration response becomes increasingly complex. Interactions among multiple damaged teeth, nonlinear tooth contact, backlash, and structural resonances generate broadband vibration energy, irregular impact responses, and nonlinear dynamic behavior [137].

These characteristic vibration signatures form the basis for vibration-based gear fault diagnosis and guide the selection of appropriate signal-processing techniques. Accordingly, advanced signal-processing methods are commonly employed to separate overlapping fault signatures, improve diagnostic reliability, and facilitate the detection, localization, and severity assessment of gear defects [137]. Because gearbox failures can result in significant production losses, costly repairs, and unexpected downtime, early detection of these vibration signatures is essential for reliable condition monitoring throughout the service life of gear transmission systems.

#### 4.5. Bladed machine faults

Bladed machines such as fans, compressors, and turbines generate vibration primarily due to periodic blade-passing events. In these systems, aerodynamic forces act continuously on the rotating blades, and any deviation in blade condition can significantly alter the unsteady flow field and the resulting rotor dynamic response [138]. Blade-related faults, such as erosion, fouling, cracking, and loss of stiffness, alter the aerodynamic loading distribution along the rotor. These changes introduce periodic, often asymmetric excitation forces that increase vibration levels at the blade-passing frequency (BPF), defined as the product of the number of blades and the rotational speed [139].

Blade faults are typically characterized by elevated vibration at the BPF together with amplitude modulation and sideband structures arising from non-uniform aerodynamic loading, particularly in multi-stage turbomachinery [140]. As fault severity increases, the vibration response becomes progressively more complex because of increased excitation and aerodynamic asymmetry [141,142].

Blade faults are commonly diagnosed by identifying elevated vibration at the blade-passing frequency, together with modulation sidebands and changes in the associated spectral characteristics. These vibration signatures provide effective indicators for the early detection and monitoring of blade deterioration in rotating machinery [141,142].

#### 4.6. Electrical faults in motors

Electrical faults in rotating electrical machines produce measurable vibration signatures due to variations in electromagnetic forces within the motor. Although vibration analysis is traditionally associated with mechanical fault diagnosis, many electrical defects also produce characteristic vibration responses that can be captured by standard monitoring systems [142]. Because the electrical and mechanical subsystems are strongly coupled in electric motors, electrical faults can influence rotor dynamics, bearing loads, and overall vibration behavior [143].

Common electrical faults in industrial motors include stator winding defects, broken rotor bars, air-gap eccentricity, rotor winding faults, and magnetic imbalance [144]. These faults disturb the electromagnetic field distribution, generating fluctuating forces on the rotor and stator. As a result, vibration components appear at frequencies related to the electrical supply frequency, rotor speed, slip frequency, or their combinations, producing complex vibration patterns [145].

Stator winding faults typically arise from insulation degradation due to thermal aging, contamination, moisture ingress, or electrical over-stress [146]. Progressive insulation failure can lead to inter-turn or phase-to-phase short circuits, creating asymmetrical magnetic fields and unbalanced electromagnetic forces. Broken rotor bars cause non-uniform rotor current distribution and periodic electromagnetic torque pulsations, whereas air-gap eccentricity results from a non-uniform air gap caused by manufacturing tolerances, bearing wear, shaft deflection, rotor unbalance, improper assembly, or structural deformation [147–149].

Electrical faults are typically characterized by vibration components at the electrical supply frequency, rotational frequency, slip frequency, and their combinations. Stator winding faults often produce increased vibration at multiples of the electrical supply frequency owing to unbalanced electromagnetic forces. Broken rotor bars generate characteristic sideband components associated with periodic torque pulsations, while air-gap eccentricity introduces vibration components related to the rotational speed, supply frequency, and their interactions [145–149]. Because electrical and mechanical phenomena are strongly coupled, these vibration signatures are frequently superimposed on those produced by mechanical faults, resulting in complex spectral patterns [150].

Electrical faults are commonly diagnosed using vibration spectrum analysis to identify characteristic electromagnetic frequency components and their sidebands. However, because electrical fault signatures

often overlap with those of mechanical faults, reliable diagnosis generally requires integrating vibration analysis with complementary techniques such as motor current signature analysis, electrical power monitoring, and thermal measurements [151]. This multi-sensor approach improves diagnostic reliability by enabling the simultaneous assessment of electrical and mechanical degradation mechanisms.

The fault mechanisms discussed in this section generate distinct vibration features that form the basis of vibration-based condition monitoring and fault diagnosis. Effective fault identification, however, requires appropriate signal processing techniques to extract and interpret these features. Table 5 summarizes the relationships among common fault types, their characteristic vibration features, and the signal processing techniques commonly used for their analysis. The techniques presented in Table 5 are discussed in detail in Section 5.

### 5. Signal processing techniques and intelligent fault diagnosis frameworks

This section reviews the principal signal processing techniques and intelligent diagnostic frameworks used for vibration-based condition monitoring of rotating machinery, addressing RQ3.

#### 5.1. Signal processing techniques

##### 5.1.1. Time-domain analysis

Time-domain analysis is the most fundamental approach to vibration signal evaluation, as it examines vibration amplitude directly as a function of time, without transforming it into other domains [152]. Since vibration data are inherently acquired in the time domain, the resulting waveform reflects the machine's instantaneous dynamic response and forms the basis of VBCM.

As an initial diagnostic step, time-domain analysis provides a rapid assessment of machine condition by revealing changes in vibration severity, impulsive events, amplitude modulation, and non-stationary behavior [153]. These characteristics are directly linked to fault mechanisms discussed in Section 3. For instance, bearing and gear defects typically produce impulsive responses, rotor unbalance increases overall amplitude, while misalignment and looseness often introduce waveform distortions [75,114].

Diagnostic evaluation relies on both waveform interpretation and statistical feature extraction. Under healthy conditions, vibration signals are typically stable and quasi-periodic. As faults develop, signals exhibit increased amplitude, irregular fluctuations, and transient impacts [152]. These variations provide valuable information for detecting abnormal operating conditions and monitoring fault progression. Common statistical indicators used to quantify these changes are RMS, peak amplitude, peak-to-peak amplitude, crest factor, and kurtosis [154]. Figure 17 shows a time-domain vibration waveform.

Among the most widely used time-domain indicators is the RMS value, which is the effective energy content of the vibration signal and is commonly used as an overall measure of machine vibration severity. It is defined as [153]:

$$A_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2} \quad (8)$$

where  $N$  is the number of samples and  $x_i$  is the instantaneous vibration amplitude. Because RMS incorporates the contribution of all vibration components within the measured bandwidth, it provides a reliable indicator for condition trending, alarm generation, and comparison with vibration severity limits specified in international standards [155]. In industrial practice, RMS vibration velocity is widely used as a primary indicator of overall machine health.

Peak and peak-to-peak amplitudes quantify the maximum instantaneous vibration response and are highly sensitive to transient events and

**Table 5**

Mapping between common fault types, characteristic vibration features, and commonly used signal processing techniques.

| Fault type                          | Characteristic vibration features                                                                                                                                       | Signal processing techniques                                                       | Ref.      |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------|
| Rotor unbalance                     | Dominant $1 \times$ running-speed component, elevated radial vibration, low harmonic content, and stable phase                                                          | Time-domain analysis, FFT, and phase analysis                                      | [71,74]   |
| Shaft misalignment                  | Strong $1 \times$ and $2 \times$ components, higher harmonics with increasing severity, elevated axial vibration, and abnormal phase across the coupling                | FFT, phase analysis, and time-frequency analysis                                   | [80–82]   |
| Bent shaft                          | Dominant $1 \times$ component, elevated axial vibration, complex phase behavior, and sensitivity to temperature and load                                                | FFT, phase analysis, orbit analysis, and time-frequency analysis                   | [85–87]   |
| Mechanical looseness                | Multiple running-speed harmonics, possible subharmonics, impact transients, amplitude modulation, and unstable amplitudes                                               | Time-domain analysis, FFT, wavelet analysis, and adaptive signal decomposition     | [90,91]   |
| Oil whirl and oil whip              | Subsynchronous components; oil whirl generally occurs around $0.4 \times - 0.5 \times$ running speed, whereas oil whip becomes locked near a rotor natural frequency    | FFT, order analysis, orbit analysis, and time-frequency analysis                   | [98–100]  |
| Rolling-element bearing defects     | Periodic impulsive responses and characteristic defect frequencies (BPFO, BPFI, BSF, and FTF), often accompanied by high-frequency resonance                            | Envelope analysis, wavelet analysis, FFT, and adaptive signal decomposition        | [108,109] |
| Gear faults                         | Gear-mesh frequency and harmonics, shaft-frequency sidebands, amplitude/frequency modulation, and periodic impact transients                                            | FFT, envelope analysis, time-frequency analysis, and adaptive signal decomposition | [131,132] |
| Cavitation and fluid-induced faults | Broadband and high-frequency vibration, random pressure-induced fluctuations, blade-pass components, and non-stationary behavior                                        | FFT, time-frequency analysis, wavelet analysis, and adaptive signal decomposition  | [137–139] |
| Electrical faults                   | Components at the electrical supply frequency and its multiples, rotational-frequency combinations, and sidebands related to torque pulsations and air-gap eccentricity | FFT, time-frequency analysis, and motor-current analysis                           | [145,149] |

localized impacts. The peak amplitude ( $A_p$ ) is defined as [153]:

$$A_p = \max(|x_i|) \quad (9)$$

where  $x_i$  is the instantaneous vibration amplitude. The peak-to-peak amplitude ( $A_{pp}$ ), defined as the difference between the maximum and minimum amplitudes, is expressed as:

$$A_{pp} = A_{max} - A_{min} \quad (10)$$

Unlike RMS, peak-based parameters are more sensitive to short-duration transient events and localized impacts associated with bearing defects, gear tooth damage, rotor rub, and mechanical looseness. Increases in peak amplitude can provide an early indication of impact-related faults.

Another important diagnostic parameter is the crest factor (CF), defined as the ratio of the peak amplitude to the RMS value and expressed as [153]:

$$CF = \frac{A_p}{A_{RMS}} \quad (11)$$

The crest factor quantifies the spikiness of a vibration signal. Because localized defects generate high-amplitude impacts while contributing relatively little to the overall vibration energy, the crest factor is particularly sensitive to incipient bearing and gear faults. An increasing crest factor often indicates the emergence of impulsive vibration behavior even when RMS values remain nearly unchanged [156].

Among higher-order statistical parameters, kurtosis is one of the most widely used because it quantifies the impulsiveness, or equivalently, the tailedness, of the vibration signal distribution. It is commonly defined as the standardized fourth central moment [153]:

$$KUR = \frac{\mu_4}{\sigma^4} \quad (12)$$

where  $\mu_4$  is the fourth central moment, and  $\sigma$  is the standard deviation. Under healthy operating conditions, kurtosis values are typically close to those associated with Gaussian random processes [157]. As localized defects develop, impulsive events become more frequent and pronounced, causing the vibration signal to deviate from a Gaussian distribution and resulting in higher kurtosis values. Kurtosis is particularly sensitive to early-stage bearing defects, gear faults, and other impact-generating fault mechanisms [158].

Additional statistical measures, such as skewness, variance, standard deviation, shape factor, impulse factor, and clearance factor, are also adopted to characterize vibration signal distributions and detect deviations from normal operating behavior [159]. These statistical features are frequently used as inputs to AI algorithms for automated fault classification and condition monitoring.

Although time-domain analysis provides valuable information on vibration severity, transient behavior, and fault progression, its ability to distinguish specific fault mechanisms is inherently limited because different faults can produce similar overall vibration levels. For example, rotor unbalance, shaft misalignment, and mechanical looseness all increase the RMS vibration level while exhibiting distinct characteristic frequencies [160]. Time-domain analysis is typically complemented by frequency-domain analysis, which decomposes the vibration signal into its constituent frequency components.

### 5.1.2. Frequency-domain analysis

Frequency-domain analysis is one of the most widely used techniques in VBCM because it enables the identification of periodic vibration components associated with specific machine elements and fault mechanisms [161]. Unlike time-domain analysis, which describes how vibration amplitude varies with time, frequency-domain analysis reveals how vibration energy is distributed across different frequencies, thereby facilitating the identification of characteristic fault frequencies.

The transformation from the time domain to the frequency domain is most commonly performed using the Fast Fourier Transform (FFT), which converts a measured vibration waveform into a frequency spectrum [92]. The resulting spectrum represents vibration amplitude as a function of frequency and provides a concise description of the machine's dynamic behavior. Through spectral analysis, vibration components associated with shaft rotation, rolling element bearings, gear meshing, blade-passing events, electrical excitation, and structural resonances can be separated and analyzed, facilitating the identification of their underlying fault mechanisms [161].

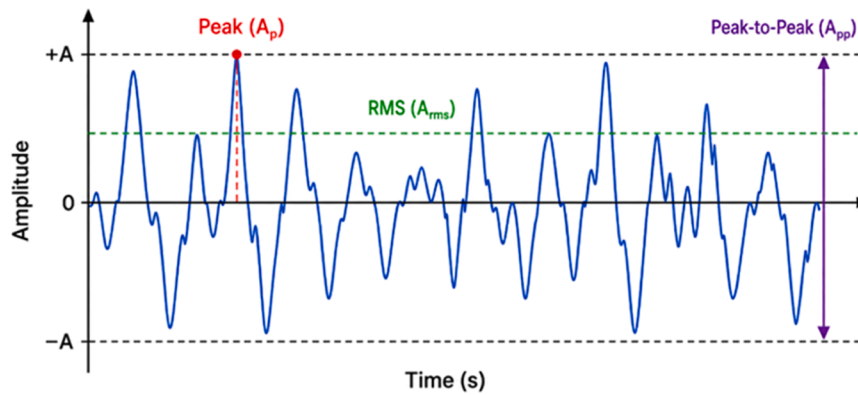


Figure 17. Time-domain vibration waveform.

The diagnostic significance of frequency-domain analysis stems from the fact that many rotating machinery faults generate vibration at frequencies determined by the machine's geometry, kinematics, and operating speed [162]. As discussed in Section 3, these characteristic frequencies serve as diagnostic fingerprints for common mechanical, fluid-induced, and electrical faults.

In addition to fault identification, frequency-domain analysis can be used to detect resonance phenomena. Resonance occurs when an excitation frequency coincides with the natural frequency of a machine component or supporting structure, resulting in dynamic amplification of the vibration response [163]. Such conditions can lead to excessive vibration, accelerated component deterioration, fatigue damage, and reduced machine reliability. Distinguishing excitation frequencies from structural resonances makes spectral analysis an essential tool for machinery troubleshooting, dynamic assessment, and resonance evaluation [164].

Modern vibration monitoring systems further enhance the effectiveness of frequency-domain analysis through automated peak identification, calculation of characteristic fault frequencies, spectral trending, and rule-based diagnostic algorithms [165]. In practical applications, spectral information is often interpreted together with phase measurements, operating conditions, machine geometry, and historical vibration trends to improve diagnostic confidence and reduce the likelihood of incorrect fault identification [166].

Despite its considerable diagnostic capability, conventional FFT analysis assumes that the vibration signal remains stationary throughout the analysis interval [20]. This assumption is generally valid for machines operating at a constant speed and load. However, under variable-speed, transient, or other non-stationary operating conditions, spectral components become smeared, distorted, or difficult to interpret, reducing the diagnostic effectiveness of conventional FFT analysis [153]. These limitations have motivated the development of advanced signal processing techniques, particularly time-frequency analysis, envelope analysis, order tracking, and adaptive signal decomposition methods, which are designed for non-stationary vibration signals and are discussed in the following subsections.

### 5.1.3. Time-frequency analysis

Time-frequency analysis is a joint representation of vibration signals in both the time and frequency domains used to characterize non-stationary behavior commonly observed in rotating machinery [167]. It is particularly useful for signals whose spectral content varies with time, such as transient events, variable-speed operation, intermittent impacts, and evolving fault conditions. In contrast to conventional frequency-domain methods, time-frequency techniques describe how frequency components evolve. This makes them more suitable for capturing dynamic changes in machine operating conditions that cannot be fully represented using FFT-based analysis [168].

The output is typically represented as a two-dimensional time-

frequency map (spectrogram), where time is plotted on the horizontal axis, frequency on the vertical axis, and signal amplitude is encoded by color intensity or contour levels [169]. This representation directly visualizes transient and time-varying vibration components that are not observable in either the time or frequency domain alone.

Several time-frequency techniques have been developed for rotating machinery diagnostics, each differing in resolution, computational complexity, and computational cost. Common methods are the Short-Time Fourier Transform (STFT), wavelet transform, Wigner-Ville distribution, Hilbert-Huang transform, and empirical mode decomposition-based approaches. Among these, the STFT is widely used for its simplicity and computational efficiency. STFT applies a sliding time window to the signal and computes the Fourier transform for each segment, producing a spectrogram that represents the local frequency content over time [170]. However, the performance of STFT is limited by the fixed window size, which introduces a trade-off between time and frequency resolution. Wavelet-based methods address this limitation by using adaptive windowing, offering higher time resolution at high frequencies and better frequency resolution at low frequencies [171]. This multi-resolution capability makes wavelet transforms particularly effective for detecting localized transient events such as bearing impacts and gear-related faults.

Time-frequency methods are particularly important in modern condition monitoring because rotating machinery often operates under variable-speed and non-stationary conditions where conventional FFT-based analysis becomes insufficient [172]. However, these techniques generally require greater computational effort and more careful interpretation than time- or frequency-domain methods [173]. The selection of an appropriate method depends on the characteristics of the signal, operating conditions, and diagnostic objectives. Nevertheless, time-frequency analysis remains a powerful approach for analyzing non-stationary vibration phenomena in rotating machinery.

### 5.1.4. Trend analysis

Trend analysis is a condition monitoring approach that evaluates the evolution of vibration features over time rather than interpreting individual measurements in isolation [18]. It reveals the long-term behavior of diagnostic indicators and enables the identification of gradual degradation processes that are not evident from single measurements. Because most faults develop progressively, trend analysis forms an essential component of condition-based and predictive maintenance strategies [174].

In practical vibration monitoring systems, measurements are acquired periodically or continuously and converted into representative diagnostic features, such as RMS value, peak amplitude, crest factor, kurtosis, and selected fault-related spectral components [154]. These parameters are stored as time-series data and tracked to identify deviations from normal operating behavior. Rather than relying exclusively on predefined alarm thresholds, trend analysis evaluates changes

relative to the machine's historical baseline, making it more robust to variations in machine design, operating conditions, and installation characteristics.

Beyond early fault detection, trend analysis provides valuable information regarding the rate of machine degradation [175]. The slope and pattern of the trend curve can be used to distinguish gradual wear processes from rapidly developing faults and to assess the urgency of maintenance actions. For example, a slow increase in vibration levels indicates normal wear or aging, whereas a rapid increase is often associated with severe defects such as bearing spalling, gear damage, lubrication failure, or the progression of mechanical looseness [176].

In industrial practice, trend-based condition assessment is frequently integrated with alarm criteria and vibration severity limits specified in international standards such as ISO 20816 [177]. Combining historical vibration trends with standardized evaluation criteria enables more reliable maintenance decision-making and reduces the risk of unexpected failures. Although trend analysis is not intended to directly identify specific fault types, it provides valuable insights into the evolution of machine health and degradation processes. Consequently, it is widely employed in conjunction with time-domain, frequency-domain, and advanced signal processing techniques to enhance diagnostic reliability and to support predictive maintenance, fault prognosis, and remaining useful life (RUL) estimation [178].

#### 5.1.5. Envelope analysis

Envelope analysis is an advanced signal processing technique specifically developed for extracting periodic impulsive signatures generated by localized defects in rolling element bearings and gear systems [179]. Defects such as surface spalls, cracks, and localized pitting generate short-duration impact forces when rolling elements or gear teeth interact with damaged surfaces [180]. These impacts excite the machine's structural resonances, producing high-frequency vibration components that are often masked by background noise and other vibration sources in the raw time-domain signal.

The principle of envelope analysis is shown in Figure 18. The raw

vibration signal contains periodic impulsive components superimposed on other vibration sources and background noise, making the fault-related information difficult to identify directly [181]. The signal is first band-pass filtered around a resonance frequency band to isolate the fault-induced high-frequency resonant response. The envelope of the filtered signal is then extracted via amplitude demodulation, and its frequency spectrum is computed to reveal the characteristic fault frequencies and their harmonics.

The envelope analysis process is based on amplitude demodulation, in which high-frequency resonant vibrations serve as a carrier signal modulated by low-frequency fault-repetition events [110]. This demodulation process enables the extraction of periodic impulsive signatures that are otherwise difficult to detect directly from the raw vibration signal. In practice, various demodulation techniques, such as rectification and the Hilbert transform, are employed to obtain the signal envelope, whose frequency spectrum reveals characteristic fault frequencies and the harmonics associated with localized defects [182].

The diagnostic performance of envelope analysis strongly depends on the appropriate selection of the frequency band, as improper filtering can suppress fault-related information or introduce spurious spectral components [183,184]. When the resonance band is properly selected, envelope analysis exhibits high sensitivity to weak and incipient defects, making it one of the most widely adopted approaches for the early detection of rolling element bearing and gear faults in industrial condition monitoring systems [185]. As a result, envelope analysis has become a crucial tool in modern VBCM and fault diagnosis.

#### 5.1.6. Order tracking analysis

Order tracking analysis is a specialized vibration analysis technique designed for rotating machinery operating under variable-speed conditions, where conventional frequency-domain methods such as the FFT fail to provide clear diagnostic information. FFT-based analysis assumes stationary operating conditions; however, in many practical applications, rotational speed varies continuously, leading to spectral smearing, peak broadening, and reduced diagnostic clarity [186]. To overcome

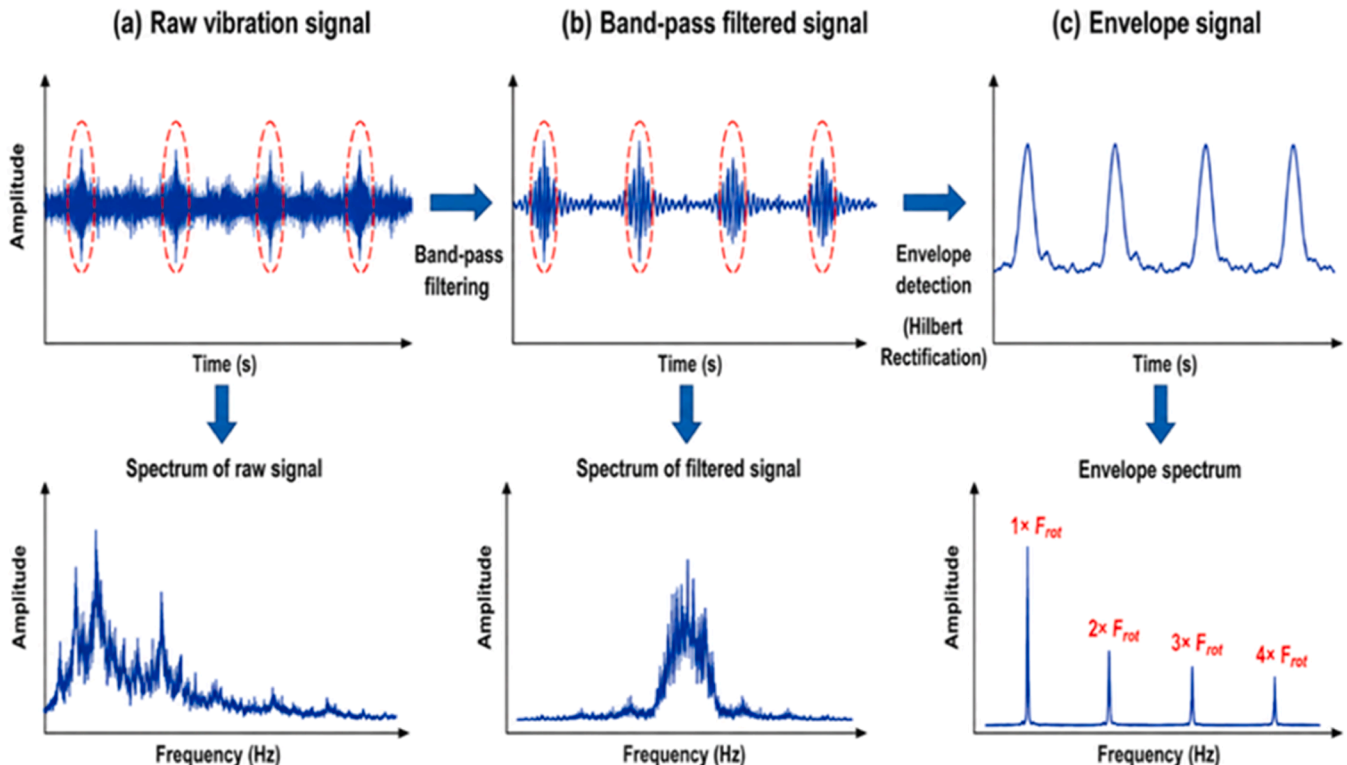


Figure 18. Principle of envelope analysis for extracting fault characteristic frequencies from impulsive vibration signals in rotating machinery.

these limitations, order tracking transforms vibration signals from the time domain into the angular domain, where vibration components are expressed as multiples of the rotational speed rather than as absolute frequencies [38]. These multiples, referred to as orders, represent machine-speed-normalized vibration components, with the first order corresponding to once-per-revolution phenomena, the second order to twice-per-revolution components, and higher or fractional orders reflecting more complex dynamic interactions. As a result, order tracking preserves the coherence of speed-dependent components and enables more reliable fault diagnosis under non-stationary operating conditions.

In practical applications, order tracking is commonly implemented using resampling techniques based on tachometer or encoder signals, whereby the time-domain vibration signal is converted into an angular-domain representation prior to spectral analysis [187]. Alternatively, order-tracking methods can estimate order components without explicit resampling using advanced signal processing algorithms [188].

The main advantage of order tracking lies in its ability to preserve the coherence of speed-dependent vibration components, ensuring that diagnostic features remain stable across varying operating conditions [189]. It is often integrated with conventional spectral, time-frequency, and envelope analyses to enhance fault detection and diagnosis in non-stationary rotating machinery systems.

#### 5.1.7. Adaptive signal decomposition methods

In recent years, adaptive signal decomposition methods have emerged as powerful tools for analyzing non-stationary and nonlinear vibration signals in rotating machinery, particularly under complex operating conditions where conventional Fourier-based techniques are less effective [190]. These methods adaptively decompose a signal into a set of intrinsic oscillatory components that represent different time scales and frequency bands, thereby enabling improved extraction of fault-related information.

Among the most widely used techniques are Empirical Mode Decomposition (EMD), Ensemble Empirical Mode Decomposition (EEMD), Variational Mode Decomposition (VMD), and Local Mean Decomposition (LMD). EMD decomposes a signal into intrinsic mode functions (IMFs) based on local extrema, making it suitable for nonlinear and non-stationary vibration analysis [191,192]. However, EMD is

susceptible to mode mixing, where different frequency components are embedded within the same IMF. To mitigate this limitation, EEMD introduces white noise-assisted decomposition to improve mode separation and enhance stability [193]. More recently, VMD has gained significant attention by formulating signal decomposition as a constrained optimization problem, producing modes with well-defined frequency bandwidths and improved robustness to noise [194]. Similarly, LMD decomposes vibration signals into product functions that contain instantaneous amplitude and frequency information, making it effective for characterizing transient and impulsive faults [195]. These decomposition techniques are widely integrated with envelope analysis, time-frequency analysis, and machine learning algorithms in modern VBCM systems to improve fault diagnosis performance, prognostic analysis, and RUL estimation [196].

The signal processing techniques discussed in the preceding subsections differ in their characteristics and suitability for different vibration signal types and operating conditions. The selection of an appropriate technique depends on factors such as signal stationarity, transient behavior, noise characteristics, computational requirements, and the intended analysis objective. Table 6 provides an engineering comparison of the commonly used vibration signal processing techniques, including their representative methods, principal assumptions, and typical limitations, to facilitate their selection for vibration-based condition monitoring applications.

As shown in Table 6, the signal processing techniques differ in their characteristics, underlying assumptions, and limitations, as well as their suitability for different vibration signal types and operating conditions. No single technique is universally applicable to all vibration analysis tasks. Accordingly, the selection of an appropriate technique should be based on factors such as signal stationarity, transient behavior, noise characteristics, computational requirements, and the intended analysis objective.

Figure 19 presents a generalized decision framework to support the selection of suitable vibration signal processing techniques. The framework should be regarded as a practical guideline rather than a prescriptive decision tree, and should be used together with engineering judgment and machine-specific operating conditions.

**Table 6**  
Engineering comparison of vibration signal processing techniques.

| Signal processing technique                                                               | Suitable signal type                             | Noise / operating characteristics                                     | Main assumptions / limitations                                                                              | Computational cost                                                            | Ref.           |
|-------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------|
| Time-domain analysis (RMS, peak, crest factor, kurtosis)                                  | General vibration signals                        | Sensitive to measurement noise and operating-condition variations     | Limited fault discrimination because different faults can produce similar statistical indicators            | Simple feature extraction and analysis                                        | [19,150, 160]  |
| Frequency-domain analysis (FFT, cepstral analysis, autoregressive methods)                | Stationary signals                               | Performance degrades under noisy and variable-speed conditions        | Assumes signal stationarity; spectral leakage may occur                                                     | Efficient spectral analysis                                                   | [20,153]       |
| Time-frequency analysis (STFT, Wavelet, Synchrosqueezing Transform)                       | Non-stationary and transient signals             | Better suited for transient and time-varying signals than FFT         | STFT has fixed time-frequency resolution; wavelet-based methods provide multi-resolution analysis           | Time-frequency representation requiring greater computational effort than FFT | [167,170, 171] |
| Trend analysis                                                                            | Long-term vibration monitoring                   | Requires measurements obtained under comparable operating conditions  | Indicates degradation trends but does not directly identify fault types                                     | Continuous or periodic feature trending                                       | [174,176]      |
| Envelope analysis (Hilbert envelope, spectral kurtosis, kurtogram, sparse representation) | Periodic impulsive signals (bearing/gear faults) | Depends on resonance-band selection and signal quality                | Incorrect resonance-band selection may suppress fault-related information                                   | Demodulation-based analysis                                                   | [179,182, 185] |
| Order tracking analysis (Order tracking, angular resampling, time-synchronous averaging)  | Variable-speed rotating machinery                | Requires accurate tachometer/keyphasor or rotational-speed estimation | Preserves speed-dependent components; inaccurate speed estimation or resampling reduces diagnostic accuracy | Angular-domain signal processing                                              | [186–189]      |
| Adaptive signal decomposition (EMD, EEMD, VMD, LMD)                                       | Nonlinear and non-stationary signals             | Performance depends on decomposition quality and algorithm selection  | Mode mixing may occur in EMD; decomposition performance depends on algorithm and parameter settings         | Adaptive iterative decomposition                                              | [191–196]      |

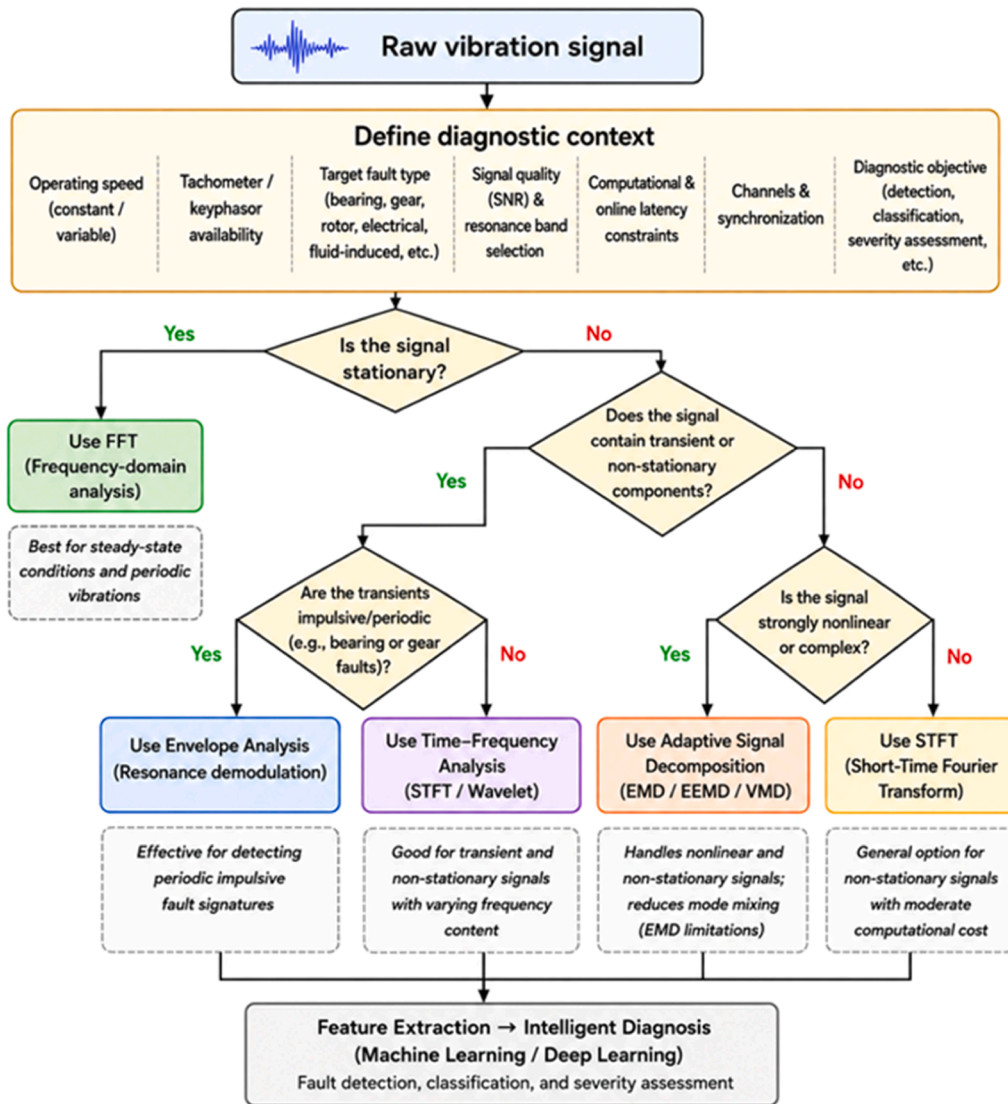


Figure 19. Decision framework for selecting vibration signal processing techniques.

## 5.2. Intelligent fault diagnosis frameworks

In recent years, AI has become increasingly important in vibration-based fault diagnosis due to its ability to automatically identify complex patterns in vibration data without requiring explicit analytical models of machine behavior [3,197]. AI has emerged as a powerful tool for condition monitoring, particularly under complex, nonlinear, and non-stationary operating conditions where conventional approaches often face limitations [198].

### 5.2.1. Conventional machine learning-based diagnosis

In rotating machinery applications, the fault diagnosis problem is typically formulated as a classification or pattern recognition task, in which vibration signals are mapped to predefined machine health states, such as normal, unbalance, misalignment, bearing defects, and gear faults [199]. The general workflow of the (ML) fault diagnosis is shown in Figure 20. The process begins with vibration data acquisition, followed by preprocessing steps such as filtering, normalization,

segmentation, and denoising. Feature extraction is then performed using either manually engineered features derived from time, frequency, and time-frequency analyses or automated representation learning techniques. Finally, the extracted features are used to train an ML model for fault classification and condition assessment.

The effectiveness of ML models largely depends on the quality of the extracted vibration features [200]. Commonly used features are statistical indicators such as RMS, kurtosis, skewness, and crest factor, as well as spectral energy features computed within selected frequency bands. These features are derived based on prior knowledge of rotating machinery dynamics and fault signatures. Once extracted, they are used as inputs to supervised learning algorithms such as support vector machines, k-nearest neighbors, decision trees, and random forests [201]. These models learn decision boundaries in the feature space that separate different operating conditions for machines. However, these ML approaches often struggle with raw or high-dimensional vibration data, especially under variable-speed, noisy, and evolving fault conditions [202].

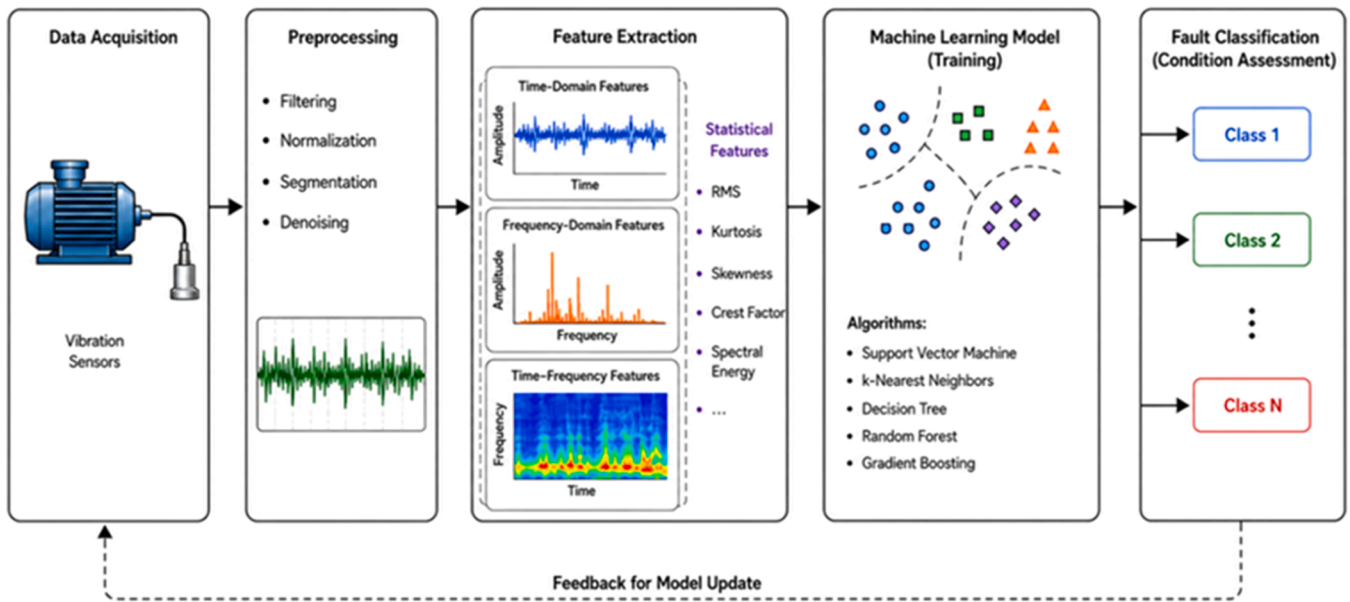


Figure 20. ML-based fault diagnosis workflow for rotating machinery.

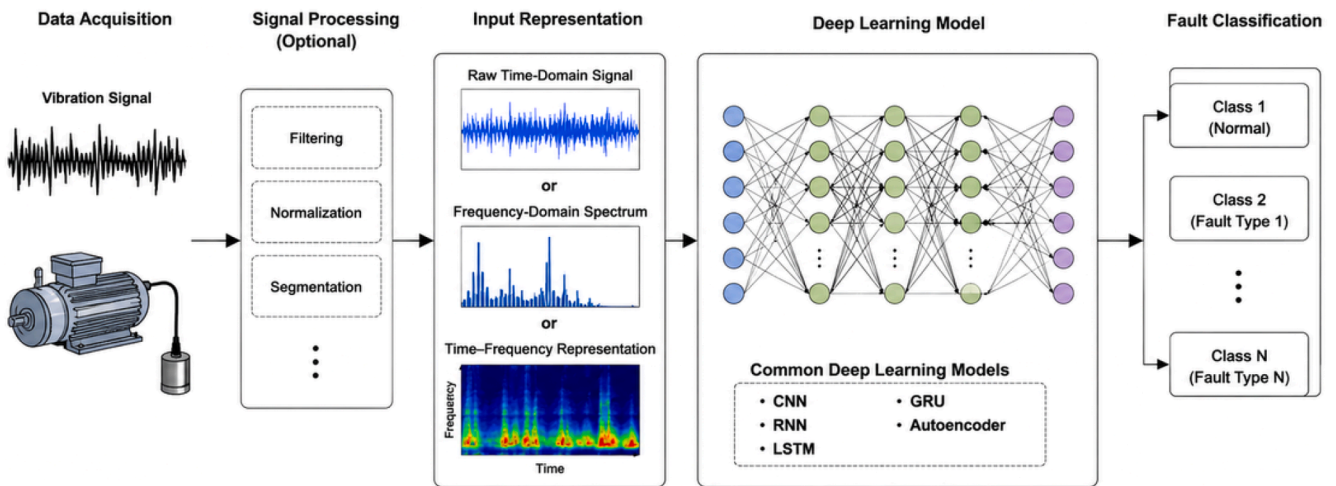


Figure 21. DL-based fault diagnosis framework.

### 5.2.2. Deep learning-based diagnosis

Deep learning (DL) methods have gained significant attention due to their ability to automatically learn hierarchical feature representations directly from raw vibration signals or transformed inputs, without requiring manual feature engineering. Unlike conventional approaches that rely on handcrafted feature extraction, DL models can directly learn discriminative patterns from data representations in either raw one-dimensional signals or processed forms obtained through standard signal processing techniques [203]. As shown in Figure 21, DL methods eliminate the need for manual feature design while improving robustness under complex, nonlinear, and non-stationary operating conditions.

Common deep learning architectures used in vibration-based fault diagnosis are Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), and autoencoders. These models are widely adopted for their ability to capture spatial patterns, temporal dependencies, and latent representations in vibration data, enabling accurate fault classification and condition assessment. However, DL

models typically require large labeled datasets for effective training, which remains a major limitation in industrial applications where fault data are scarce [204]. In addition, certain fault types, such as broken rotor bars or evolving eccentricity, are difficult to reproduce experimentally without risking machine damage. More critically, models trained on a specific machine often fail to generalize to other machines due to differences in structural dynamics, power ratings, and operating conditions [205]. Another important challenge is interpretability: DL models achieve high accuracy, yet their decision-making processes remain largely opaque.

### 5.2.3. Emerging intelligent and hybrid methods

Recent advances in machine condition monitoring have led to the development of emerging intelligent and hybrid diagnostic approaches that address the limitations of conventional ML and DL methods. These limitations are dependence on large labeled datasets, poor generalization across different machines and operating conditions, and limited interpretability of model decisions. To overcome these challenges, certain advanced learning strategies have been introduced. For example,

transfer learning and domain adaptation techniques have been developed to mitigate the problem of distribution mismatch between training and deployment environments [206,207]. These methods transfer knowledge learned from a source domain, such as laboratory-generated datasets or data from similar machines, to a target domain with limited or no labeled data. As a result, transfer learning and domain adaptation significantly reduce the requirement for extensive fault labeling while improving cross-machine and cross-condition generalization.

Self-supervised and unsupervised learning approaches further reduce the dependency on labeled data by learning meaningful feature representations directly from unlabeled vibration signals [208]. These methods exploit the inherent structure of vibration data to extract informative patterns, making them particularly suitable for industrial applications in which fault labels are scarce, incomplete, or unavailable.

Explainable artificial intelligence (XAI) techniques enhance the transparency and interpretability of diagnostic models by identifying the contribution of input features or signal regions to the final decision [209]. This improves user trust and supports more reliable decision-making in practical condition-monitoring systems, particularly in safety-critical industrial environments.

Finally, hybrid physics-data-driven models integrate domain knowledge from vibration theory and machine dynamics with data-driven learning techniques [210]. By combining physical interpretability with the adaptability of AI models, these approaches improve robustness, generalization, and diagnostic reliability. Thus, hybrid models are the most promising directions for next-generation intelligent fault diagnosis systems.

The intelligent diagnostic methods discussed above differ in their data requirements, model characteristics, practical applicability, and limitations. Table 7 summarizes the main characteristics of the commonly used intelligent diagnosis methods.

As presented in Table 7, the applicability and limitations of intelligent diagnostic methods vary according to data availability, operating conditions, computational requirements, and interpretability. Conventional machine-learning algorithms are generally suitable when handcrafted features and relatively limited datasets are available, whereas deep-learning methods are advantageous for large-scale vibration datasets because they automatically learn discriminative features. Transfer learning is particularly useful when labeled fault data are limited or operating conditions differ between source and target machines, whereas domain adaptation improves diagnostic performance under distribution shifts caused by changes in machine type or operating conditions. Despite these advances, challenges remain regarding model interpretability, generalization, computational cost, and the availability of representative labeled datasets.

### 5.3. Representative workflow for bearing fault diagnosis

Figure 22 presents a representative conceptual workflow for vibration-based bearing fault diagnosis using vibration signals. Rather than describing a specific experimental case study, the figure

summarizes the typical diagnostic procedure commonly reported in the literature. The workflow begins with vibration data acquisition using an accelerometer mounted on the bearing housing. The acquired signal is subsequently preprocessed through filtering, segmentation, and normalization to improve signal quality. Diagnostic features are then extracted using complementary time-domain, frequency-domain, and time-frequency analyses before being interpreted using conventional signal-processing techniques or AI-based diagnostic models to identify the fault type and assess the machine condition. The specific pre-processing parameters, signal-processing settings, diagnostic models, and evaluation procedures depend on the machine configuration, measurement conditions, and application requirements, and are therefore beyond the scope of this representative workflow.

In a typical bearing fault diagnosis scenario, localized defects on the inner race, outer race, or rolling elements generate repetitive impulsive vibration responses as the rolling elements pass over the damaged regions. These impacts appear as impulsive waveforms in the time domain, characteristic defect frequencies and their harmonics in the frequency spectrum, and localized transient energy in the time-frequency representation. The complementary interpretation of these signal representations improves diagnostic reliability and supports accurate fault identification and condition assessment.

## 6. Standards, industrial implementation, and emerging trends

This section presents international standards, industrial implementation practices, and emerging trends in vibration-based condition monitoring, addressing RQ4 and RQ5.

### 6.1. Industrial standards for vibration monitoring

#### 6.1.1. API standards for vibration monitoring

The American Petroleum Institute (API), established in 1919, is a leading standards-development organization for the oil and gas industry. It has created more than 800 consensus-based standards addressing safety, reliability, and operational performance [211]. Today, many API standards are widely used in petrochemical, power generation, and other process industries [212].

Among the standards related to vibration, API 670 (Machinery Protection Systems) is the most widely used framework for monitoring and protection of critical rotating machinery [213]. The main feature of API 670 is its focus on vibration measurements taken from the shaft using eddy-current proximity probes installed near the rotor shaft [214]. The standard also specifies requirements for sensor installation, calibration, linearity, operating range, and system verification.

In addition to API 670, other API standards support vibration monitoring for specific uses. API 678 (Accelerometer-Based Vibration Monitoring System) was developed as a purchasing standard for casing vibration monitoring instrumentation. However, due to its limited scope and the absence of vibration severity limits, its provisions were later incorporated into the third edition of API 670. Similarly, API 613 and

**Table 7**  
Comparison of commonly used intelligent fault diagnosis methods.

| Diagnosis method                  | Data requirement                                     | Model characteristic                                               | Main limitation                                                 | Ref.      |
|-----------------------------------|------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------|-----------|
| Conventional machine learning     | Handcrafted features and labeled datasets            | Relies on manually engineered features for classification          | Performance depends on feature engineering and feature quality  | [199–201] |
| Deep learning                     | Large labeled datasets                               | Learns feature representations directly from raw or processed data | High computational demand and limited model transparency        | [203–205] |
| Transfer learning                 | Source-domain data with limited target-domain labels | Transfers learned knowledge between related domains                | Performance depends on source-target domain similarity          | [206,207] |
| Self-/unsupervised learning       | Primarily unlabeled vibration data                   | Learns feature representations without extensive labeled data      | May provide lower discrimination for some complex fault classes | [208]     |
| Explainable AI                    | Trained AI models                                    | Provides explanations for model predictions                        | Depends on the underlying explanation technique                 | [209]     |
| Hybrid physics–data-driven models | Physical knowledge and operational data              | Integrates physics-based models with data-driven learning          | Increased implementation and modeling complexity                | [210]     |

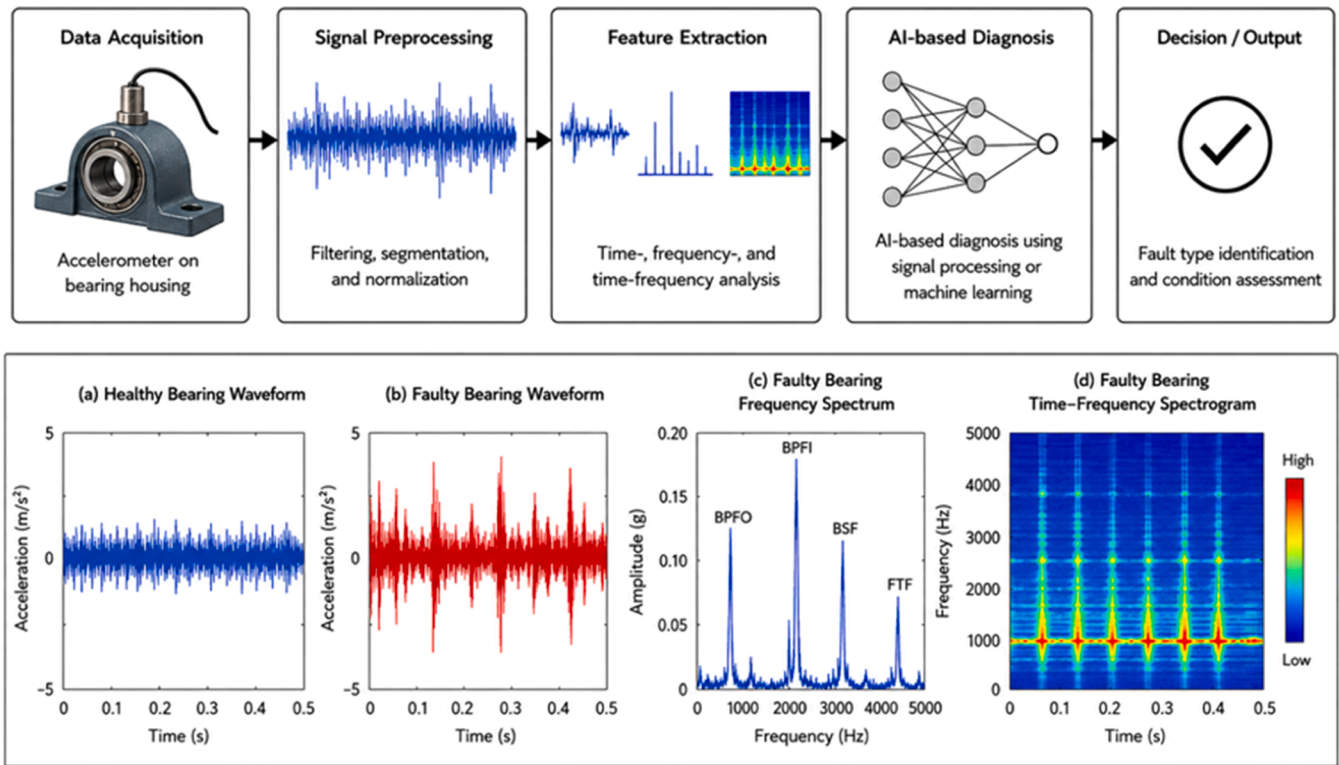


Figure 22. Representative workflow for vibration-based bearing fault diagnosis.

API 677 set acceptance limits for casing vibrations in newly commissioned special-purpose and general-purpose gear units, respectively [215]. These standards use RMS velocity and true-peak acceleration for overall and frequency-filtered measurements.

The main focus of API vibration standards is on machinery protection and system performance, not on defining universal vibration severity limits. Although API 670 centers on monitoring system requirements and protective functions, vibration acceptance criteria are typically addressed in equipment-specific standards. API standards are particularly useful for machinery protection and acceptance testing, but are often complemented by ISO-based condition assessment methods and advanced diagnostic techniques for ongoing condition monitoring and fault diagnosis.

#### 6.1.2. VDI and early ISO velocity-based standards

Before modern ISO standards, vibration severity assessment was primarily based on vibration velocity measurements obtained from housing-mounted sensors. In this context, VDI 2056 and ISO 2372 established some of the first widely accepted frameworks for evaluating vibration severity in rotating machinery [216,217]. These standards classified machines by size, mounting condition, and operating speed, and defined qualitative severity zones corresponding to acceptable, restricted, and unacceptable operating conditions.

In addition to these formal standards, earlier practical methods contributed to the development of vibration severity assessment. The Rathbone standard [218] introduced amplitude-versus-frequency severity curves that approximated constant velocity. Meanwhile, the Yates standard [219] was developed based on tests conducted on marine geared turbine installations. Soon after, the IRD vibration severity chart defined vibration limits based on peak vibration measured externally on machine casings [177]. These early methods provided useful reference points for evaluating machine conditions and influenced later standardization efforts.

Although later replaced by more advanced standards, VDI 2056 and ISO 2372 were instrumental in establishing the basic ideas of vibration

severity assessment. ISO 2372 has since been withdrawn and replaced by the ISO 10816 series, although it is still referenced in older monitoring systems and historical industrial documentation [220]. Similarly, VDI 2056 continues to serve as an engineering reference and is still used in some industrial sectors, particularly in Europe [221].

One major drawback of these early velocity-based standards is the reliance on overall vibration magnitude without considering frequency content or time-varying behavior. Despite this, the simplicity, robustness, and ease of implementation of these approaches contributed to the standardization of vibration-based condition assessment and laid the foundation for the more advanced ISO 10816 and ISO 20816 frameworks discussed in the next section [177].

#### 6.1.3. ISO 10816 and ISO 20816 standards

The ISO 10816 series and its successor, ISO 20816, are the most widely used international standards for evaluating vibration severity in rotating machinery. These standards provide a practical approach to machine condition assessment based on vibration measurements from non-rotating components, typically machine housings or bearing supports [222]. The ISO 10816 series defined RMS vibration velocity as the main indicator of vibration severity and set the measurement bandwidth (typically 10 to 1000 Hz) and evaluation procedures for casing vibration assessment [177]. Thus, RMS vibration velocity remains the primary parameter for evaluating the overall condition and dynamic behavior of the machine.

The main strength of the ISO 10816/20816 framework is its organized machine classification system. Machines are grouped based on factors such as rated power, shaft height, bearing type, mounting configuration, and foundation stiffness [177]. This allows vibration severity limits to be adjusted for specific machine groups rather than using a universal threshold. Within this system, vibration severity is divided into four defined condition zones: Zone A, Zone B, Zone C, and Zone D. Zone A represents newly commissioned machines or machines in excellent condition. Zone B is for machines that operate acceptably for continuous service. Zone C corresponds to conditions requiring planned

corrective maintenance, and Zone D corresponds to vibration levels that require immediate action to prevent damage.

ISO 20816 builds on the ISO 10816 framework by covering a wider range of machine types, revised measurement practices, and improved compatibility with modern digital monitoring systems. ISO 20816-9 further extends the framework for gear units by adding vibration assessment in terms of acceleration, in addition to velocity and displacement [223]. This change addresses high-frequency vibration components in geared machinery. There is a growing need for more detailed analysis of dynamic behavior in modern high-speed rotating systems.

Recent comparative studies confirm the effectiveness of ISO 20816-based evaluation criteria in various industrial settings. In particular, Delprete and Gastaldi [177] found that ISO 20816 Parts 3 and 9 provide the most consistent and reliable vibration severity assessments among the evaluated standards. These results support the continued use of ISO 20816 as a practical guide for evaluating vibration severity and monitoring machine conditions.

#### 6.1.4. GM and AGMA standards for vibration monitoring

In addition to ISO and API standards, General Motors (GM) published a detailed Vibration Standard for Machinery and Equipment in the late 1990s [177]. This standard defines casing vibration amplitude criteria across frequency bands for various equipment classes, from precision machine tools to general-purpose motors and fans. It sets acceptance limits for new and rebuilt machines for line-amplitude and band-limited vibration levels, along with progressively increasing warning and alarm thresholds compared to the new-machine baseline. The standard also defines requirements for spectral resolution in vibration measurements. For general machinery and gear units, the velocity limits are comparable, but the acceleration limits for gear units are higher than those for general machinery.

The American Gear Manufacturers Association (AGMA) standard AGMA 6000-C20 provides guidelines for measuring linear vibrations in gear units. It establishes discrete frequency-based vibration limits for acceptance testing [224]. Casing vibration limits are specified in terms of displacement, velocity, and acceleration, with threshold values varying with pitch line velocity and gear unit classification. The velocity limits generally align with the trends defined in ISO 20816. AGMA 6000-C20 mainly applies to gear units with rigid housings, whereas higher vibration amplitudes are acceptable for flexible housings or support structures when considered during the design stage [177].

### 6.2. Practical implementation challenges

In industrial applications, the effectiveness of VBCM depends on practical implementation factors that influence diagnostic accuracy, robustness, and reliability. One of the most critical challenges is sensor placement. Improper installation or suboptimal positioning can significantly reduce sensitivity to fault-related vibration signatures [47,49]. The structural transmission path between the fault source and measurement location affects signal fidelity because vibration energy can be weakened, amplified, or distorted as it travels through machine components and support structures. Additionally, mounting conditions, local structural resonances, and cross-axis sensitivity can lead to measurement errors and reduce diagnostic reliability [14,48]. Proper sensor placement based on machine geometry, structural dynamics, and fault propagation paths is crucial for effective vibration monitoring.

Environmental and operating conditions introduce additional complexity into vibration measurements [19]. External vibration sources, temperature variation, electromagnetic interference, lubrication changes, and mechanical coupling effects can distort measured signals and hide fault-related features [54]. Moreover, many rotating machines operate under variable speed and load conditions, producing non-stationary vibration behavior that complicates spectral interpretation and reduces the effectiveness of traditional FFT-based diagnostic

methods [153].

Data quality and signal integrity also pose significant challenges in long-term monitoring systems. Sensor noise, signal drift, missing data, communication interruptions, and saturation events can degrade diagnostic performance if not properly managed [61]. These issues require robust preprocessing, such as filtering, signal validation, anomaly detection, and data-cleaning procedures, to ensure that extracted features accurately represent the machine's condition [55,225].

Another important challenge is selecting alarm thresholds and ensuring reliable decision-making. Overly sensitive thresholds cause false alarms, leading to unnecessary maintenance actions, alarm fatigue, and higher operational costs. On the other hand, overly conservative thresholds delay fault detection and allow fault progression toward severe failure [177]. This issue is particularly important in standards-based monitoring, where fixed vibration severity limits do not fully capture machine-specific behavior, operating variability, or gradual degradation trends.

### 6.3. Research challenges

Building upon the practical implementation challenges discussed in Section 6.2, this section critically reviews the major methodological and research gaps identified throughout the reviewed literature. One of the most significant challenges is the limited availability of representative fault datasets obtained under real industrial operating conditions. Many intelligent diagnostic methods are developed and evaluated using laboratory datasets collected under controlled environments, whereas industrial machines operate under varying speeds, loads, environmental conditions, and degradation mechanisms [226]. Thus, diagnostic models often experience performance degradation when deployed under operating conditions that differ from those encountered during training.

Another major challenge is the limited generalization capability of intelligent diagnostic models. Changes in machine type, structural dynamics, sensor configuration, operating conditions, and measurement environments frequently produce distribution shifts between training and deployment data [227]. As a result, models that achieve high diagnostic accuracy on benchmark datasets often fail to maintain comparable performance across different machines or industrial environments without additional adaptation or retraining.

Compound-fault diagnosis remains a major research challenge because multiple fault mechanisms frequently occur simultaneously under industrial operating conditions. Interactions among mechanical, electrical, fluid-induced, and structural faults generate overlapping vibration signatures, nonlinear dynamic behavior, and modulated frequency components that complicate fault separation and localization [228]. Recent dynamic modeling studies have demonstrated that physics-based transmission and gear dynamic models provide valuable insights into fault-induced vibration mechanisms and can improve the interpretation of complex fault signatures, particularly in geared transmission systems [229,230].

Another important research challenge concerns the interpretability and reliability of intelligent diagnostic systems. Although DL models often achieve excellent diagnostic performance, the decision-making processes of these models generally lack transparency, limiting user confidence and practical adoption in safety-critical industrial applications [231].

Finally, the lack of standardized benchmarking protocols and comprehensive industrial validation remains a major challenge for the objective evaluation and comparison of diagnostic methods [232]. Variations in datasets, fault categories, evaluation metrics, and experimental protocols make consistent performance assessment difficult. In addition, broader validation under realistic industrial operating conditions is needed to improve confidence in the robustness, generalizability, and practical applicability of intelligent diagnostic frameworks.

#### 6.4. Emerging trends and future research directions

Recent advances in VBCM are driven by advancements in AI, IIoT, Industry 4.0-enabled connectivity, and digital infrastructure. These developments are transforming conventional monitoring systems into intelligent predictive maintenance platforms capable of automated decision-making in near real time. Beyond improving monitoring capabilities, these technologies also provide promising solutions to the practical implementation challenges discussed in Section 6.2 and the research challenges identified in Section 6.3.

One important trend is the growing adoption of wireless and IIoT-enabled vibration monitoring systems. These systems allow for flexible sensor deployment and continuous data acquisition without the installation constraints of wired systems. This improves scalability and reduces deployment costs, especially in large industrial sites with geographically distributed rotating assets. However, challenges related to communication reliability, bandwidth limitations, synchronization, cybersecurity, and power consumption remain active research topics, particularly for long-term continuous monitoring [61–63].

AI, particularly ML and DL, has become a major area of research in modern fault diagnosis [197,198]. These data-driven methods enable automatic feature extraction, fault classification, anomaly detection, and prognostic modeling by learning complex patterns from vibration data. Compared to conventional model-based or manually engineered feature methods, AI-based systems can improve adaptability under nonlinear, noisy, and non-stationary operating conditions. Automatic feature learning improves robustness to noisy and non-stationary vibration signals, thereby enhancing data quality and diagnostic reliability. In addition, transfer learning improves model generalization by transferring diagnostic knowledge across different machines or operating conditions where labeled fault data are limited. Despite these benefits, there are significant research challenges related to model generalization, explainability, robustness under unseen operating conditions, and the limited availability of high-quality labeled fault datasets. Recent advances have also focused on integrating physical knowledge into intelligent diagnostic models to improve interpretability, domain generalization, and cross-machine diagnostic performance. Physics-informed and physics-guided learning frameworks effectively combine data-driven learning with machine dynamics and causal knowledge, thereby improving diagnostic robustness under varying operating conditions [233,234].

The integration of edge computing, cloud platforms, and digital twin technologies is further changing vibration-monitoring systems within Industry 4.0 ecosystems. Edge computing processes signals and detects faults close to the sensor, thereby reducing communication latency and transmission overhead while supporting rapid decision-making in systems where safety is critical [225]. Cloud-based platforms provide centralized storage, remote diagnostics, and fleet-level analytics across various industrial locations. At the same time, digital twins establish dynamic virtual models of physical assets using real-time sensor data to allow machine-state estimation, fault progression analysis, and predictive maintenance planning through simulation-based evaluation. Moreover, digital twins support adaptive threshold setting and machine-specific health assessment by continuously updating virtual models using real-time operational data, thereby overcoming the limitations of fixed alarm thresholds under varying operating conditions [235]. These technologies promote more scalable, interconnected, and intelligent maintenance systems.

Multi-sensor data fusion is also gaining more attention, as vibration data are integrated with complementary measurements such as temperature, acoustic emission, motor current, and pressure signals [236]. By integrating complementary information from multiple sensing modalities, multi-sensor data fusion improves data quality, improves diagnostic reliability, and reduces the effects of noisy or incomplete measurements. This multimodal approach improves diagnostic robustness and fault discrimination, especially in complex systems where

single-sensor analysis provides incomplete or ambiguous information.

Another major area of research is RUL prediction within prognostics and health management frameworks [237]. RUL models support maintenance scheduling based on actual degradation behavior rather than fixed maintenance intervals [238]. This transforms maintenance strategies from preventive to predictive approaches. Future research, therefore, should focus on developing more reliable hybrid diagnostic frameworks that integrate physics-based dynamic models, advanced signal processing, and physics-informed AI to improve interpretability, generalization, and decision-making reliability in industrial vibration monitoring systems.

## 7. Conclusions

In this work, a comprehensive review of vibration-based condition monitoring and fault diagnosis for rotating machinery is presented. The review covers vibration fundamentals, major fault mechanisms, signal processing techniques, intelligent diagnostic frameworks, international standards, practical implementation challenges, and emerging research directions. Based on this analysis, vibration analysis remains one of the most effective and widely used methods for machinery health monitoring due to its high sensitivity to mechanical, fluid-induced, and electromechanical faults.

The review reveals that common fault types are rotor-related faults, bearing defects, gear damage, blade faults, and electrical machine abnormalities. The reviewed literature is predominantly focused on bearing and gearbox fault diagnosis, whereas comparatively fewer studies investigate fluid-induced, electrical, and compound faults. Each fault type has unique vibration signatures that can be identified using appropriate signal processing techniques. Conventional methods, such as time-domain, frequency-domain, and time-frequency analysis, still have strong diagnostic capabilities. Meanwhile, advanced signal processing methods have significantly improved feature extraction and representation under complex and variable operating conditions, although reliable diagnosis under variable-speed conditions and compound-fault scenarios remains a significant challenge.

The review also points out that the reliability of vibration-based diagnostics depends on both analytical methods and practical implementation factors, such as sensor placement, variability in operating conditions, data quality, and threshold selection. Furthermore, the evidence indicates that many AI-based diagnostic methods have been validated primarily using laboratory datasets, while large-scale industrial validation and cross-machine generalization remain comparatively limited. Recent advancements in wireless sensing, edge computing, digital twins, multi-sensor data fusion, and data-driven prognostics are transforming traditional condition monitoring systems into intelligent predictive maintenance frameworks.

Based on the detailed analysis, the review identifies major practical challenges in vibration-based condition monitoring. Common challenges are sensor placement, environmental and operating variability, data quality and signal integrity, and the selection of reliable alarm thresholds. The review also identifies emerging trends and future research directions, such as wireless and IIoT-enabled monitoring, transfer learning and domain adaptation, explainable and hybrid AI, edge computing, digital twins, multi-sensor data fusion, standardized benchmarking, and industrially validated prognostic frameworks. Future research should therefore emphasize industrial validation, physics-informed intelligent diagnosis, improved cross-machine generalization, and standardized benchmarking to accelerate the practical deployment of intelligent vibration-based diagnostic systems.

In future work, the authors plan to extend a similar review to machine prognosis and remaining useful life prediction. The authors also intend to investigate trend analysis for modeling fault progression and degradation behavior. This could improve early fault prediction and maintenance planning in rotating machinery. Such future studies are expected to improve the integration of fault diagnosis and prognostics

for more reliable and intelligent maintenance decision-making in modern industrial systems.

### Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used an AI-assisted language tool in order to improve grammar and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

### CRediT authorship contribution statement

**Fasikaw Kibrete:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Temesgen Tadesse Feisa:** Writing – original draft, Investigation, Formal analysis. **Dereje Engida Woldemichael:** Writing – review & editing, Validation, Supervision. **Bewketu Gizachew Mekonnen:** Writing – review & editing, Validation.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

No data was used for the research described in the article.

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